A Few Questions!

GenAI has become really important?

Struggling with the pace at which it is growing

Struggling to define a learning path

A Teacher’s Perspective

We are also in the same boat

My goal for 2025

Barely covered 20 percent

Hopped playlist for playlist

Got frustrated

Need for a Roadmap

My realization - I need a roadmap!

Roadmap or Map?

Spent last 30 days building a map/mental model

Worked wonders for me

Going to share it with you

The Promise!

Full picture of Gen AI

Fit any term on the map

Locate yourself

Feel literate

Target Audience

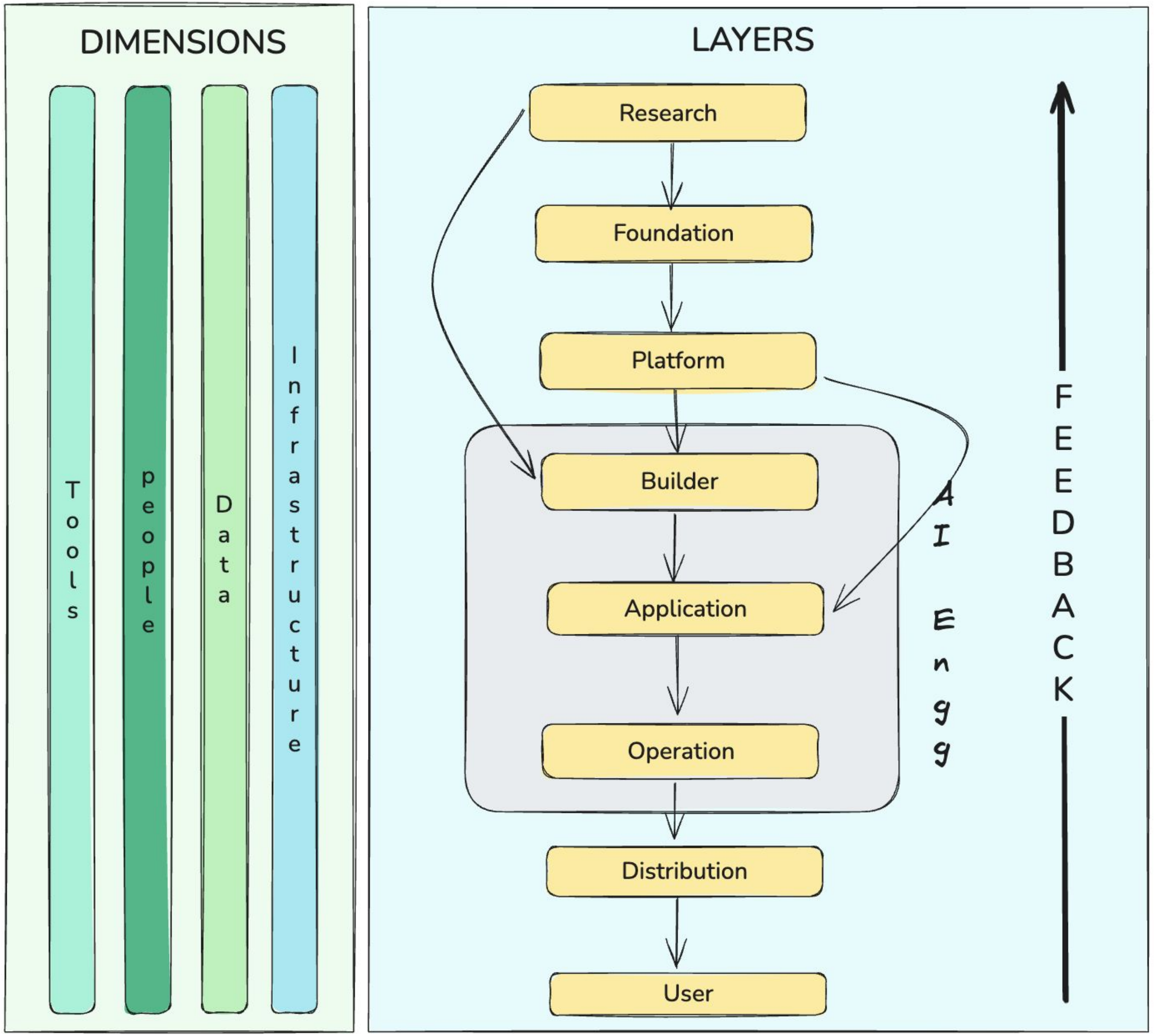
Each and everyone!

You have to just 2 things:

- Watch the entire video
- Test the map

The Map of

Generative AI



This diagram can organise the entire GenAI Landscape

Divided into 2 parts: **Layers** & **Dimensions**

RESEARCH

Research Layer

The What

The birthplace of core AI innovations

Purpose

To develop new models, learning algorithms & training techniques

Who Works Here

Researchers in Academic Institutions & Research Labs

RESEARCH

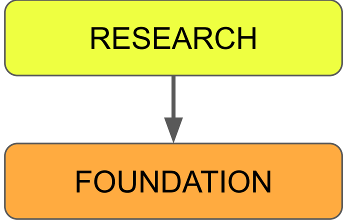
Research Layer

Focus Areas

- New Model Architectures
- Optimization Techniques
- Alignment
- Multimodality
- Interpretability
- Emergant Phenomena

Output

Research Papers, Datasets & Benchmarks



Foundation Layer

The What

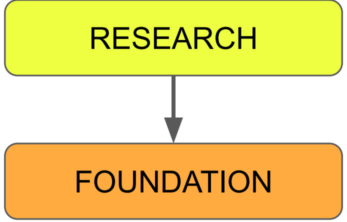
This is where research ideas are converted to large scale AI models

Purpose

To **train, align** and **fine-tune** powerful foundation models that others can use

Who Works Here

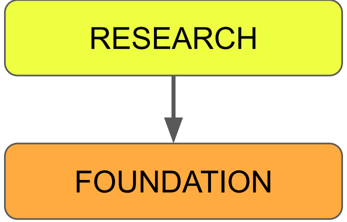
Applied Scientists, data scientists, and ML engineers in big tech labs



Foundation Layer

Focus Areas

- Data Preparation
- Model Implementation
- Distributed Training
- Evaluation
- Alignment
- Fine Tuning



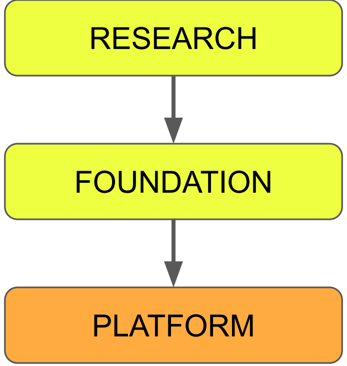
Foundation Layer

Dependencies

Research innovations, Massive datasets, Compute and Capital

Output

Base models, Aligned models & Fine tuned models



Platform Layer

The What

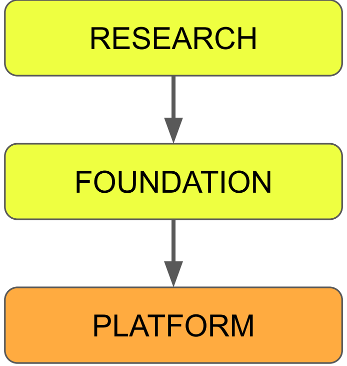
This layer brings foundation models online

Purpose

To provide easy, reliable & scalable access to Foundation models through APIs and tools.

Who Works Here

Platform engineers, API developers, MLOps specialists, cloud architects, and PMs



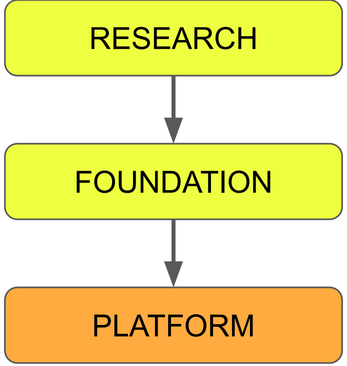
Platform Layer

Type of Foundation Model

- Proprietary Models
- Open Source Models

Delivery Modes

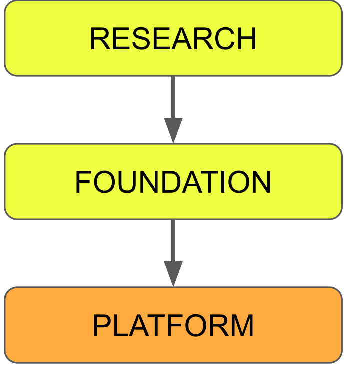
- Model APIs
- Cloud AI Platforms
- Self Hosted Models
- Hosting & Acc Services



Platform Layer

Focus Areas

- Model Serving
- Inference Optimization
- API & SDK Development
- Deployment
- Scalability Management
- Performance Monitoring



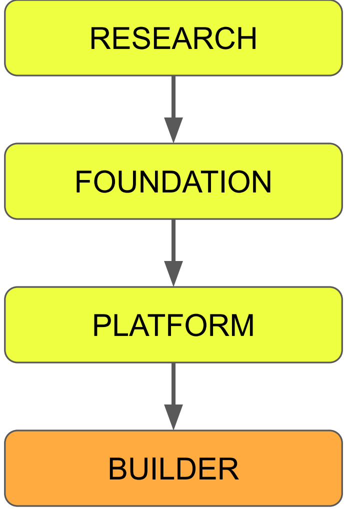
Platform Layer

Dependencies

- Trained models from the previous layer
- Infrastructure - GPU servers, network bw, storage
- Tools - Kubernetes, docker for deployment, FastAPI for development

Output

Model APIs & SDKs, tools like Ollama, services like AWS Bedrock



Builder Layer

The What

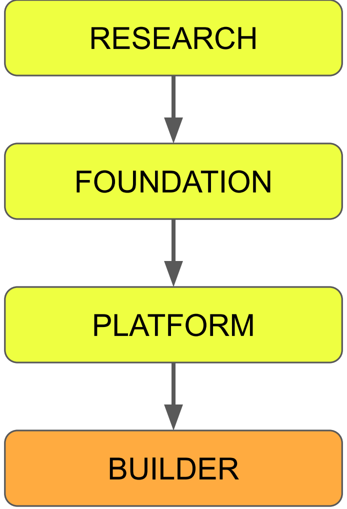
This is where intelligence takes shape into usable systems.

Purpose

To build intelligent workflows and applications by combining Models, tools and logic.

Who Works Here

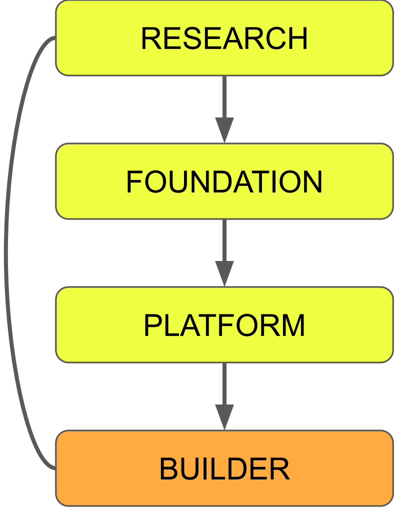
AI engineers, app developers, prompt designers, and system architects.



Builder Layer

Focus Areas

- Prompt Engineering
- RAG
- Memory Management
- Agentic AI
- Context Engineering



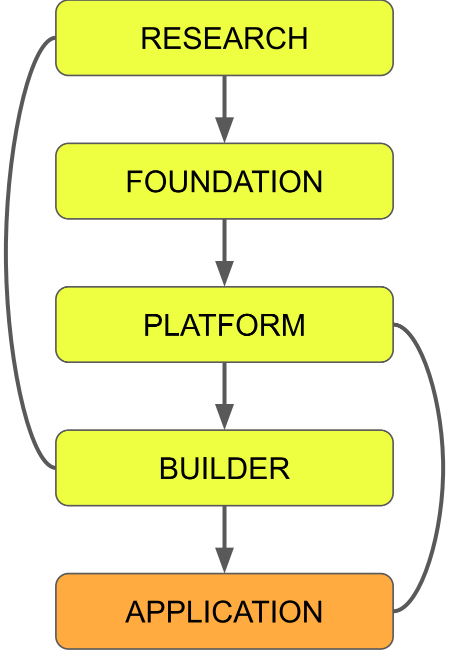
Builder Layer

Dependencies

- Model from Platform Layer
- Input from Research Layer
- Data & Context sources
- Compute & runtime

Output

- Ideas and Techniques
- Libraries, Frameworks & SDKs



Application Layer

The What

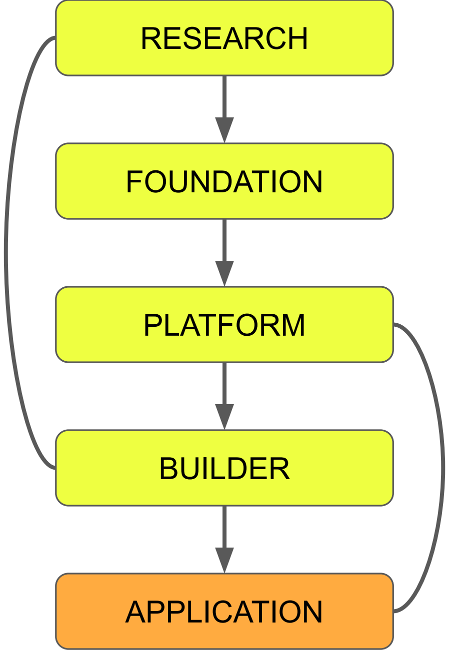
This is where AI capabilities are packaged into end user products

Purpose

To build AI native and AI integrated software for the users.

Who Works Here

AI engineers, Full stack developers, UI/UX designers, and Product managers.



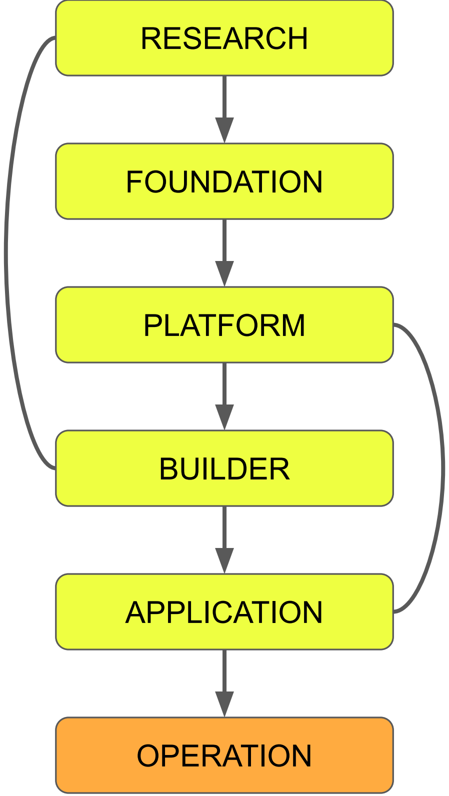
[Application Layer]

Focus Areas

- Productization
- Safety & Alignment
- Performance Optimization

Output

Production ready AI products



Operation Layer

The What

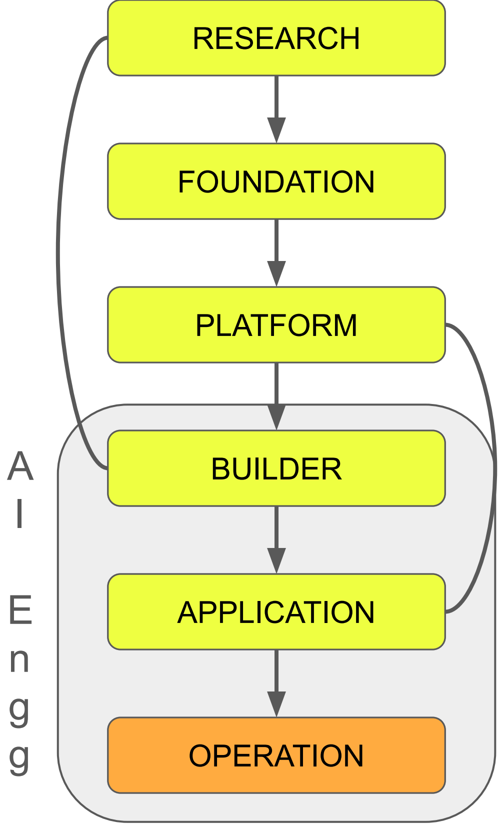
This is where AI systems are deployed, monitored, and managed in production.

Purpose

To keep AI systems stable, scalable, and continuously improving in real-world use.

Who Works Here

MLOps Engineer, DevOps Engineer, SRE, Cloud Architects, Security experts



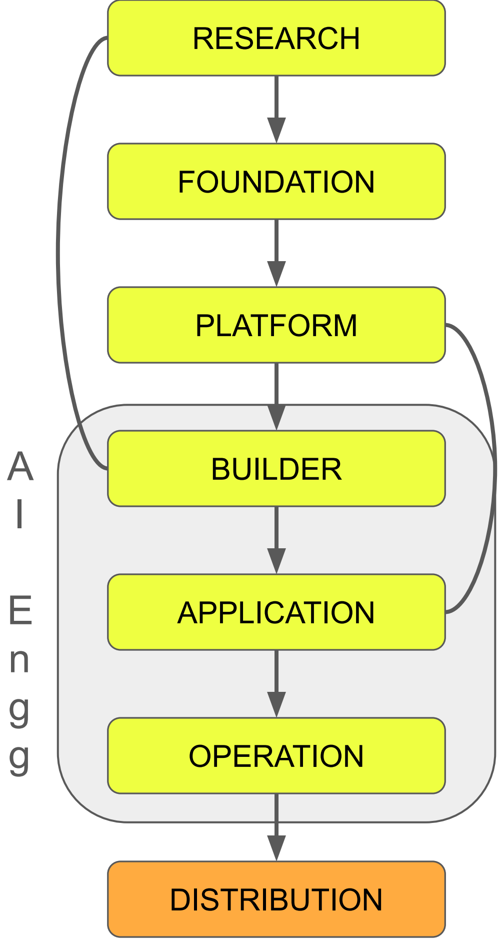
Operation Layer

Focus Areas

- Deployment
- Logging & Monitoring
- Evaluation
- Feedback

Output

Deployed AI products running reliably at scale



Distribution Layer

The What

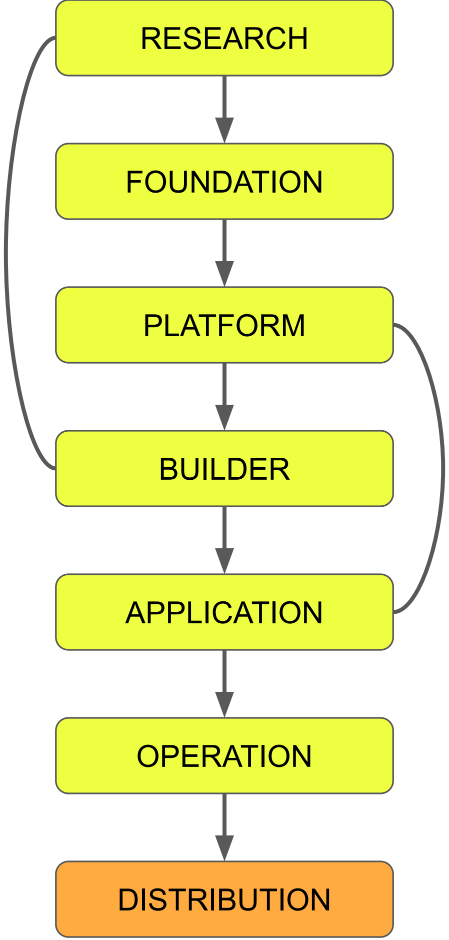
This is where AI products are reaches users through channels, platforms, and marketplaces.

Purpose

To scale the reach of AI applications by managing delivery, access, and adoption.

Who Works Here

Growth engineers, Marketing teams, product managers



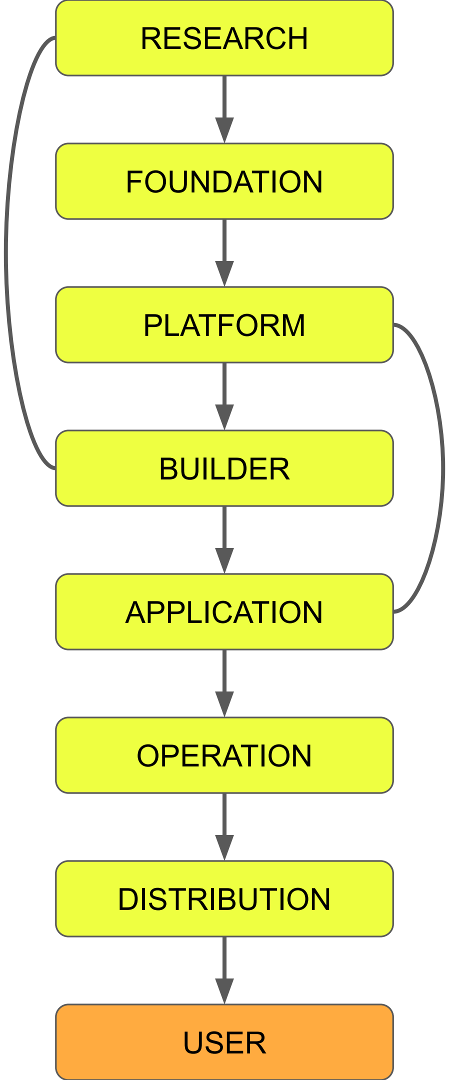
Distribution Layer

Focus Areas

- Delivery Channels
- Ecosystem Integration
- Strategic Partnerships
- Marketing & Awareness
- Compliance & Localization

Output

Market reach, user adoption and revenue growth



User Layer

The What

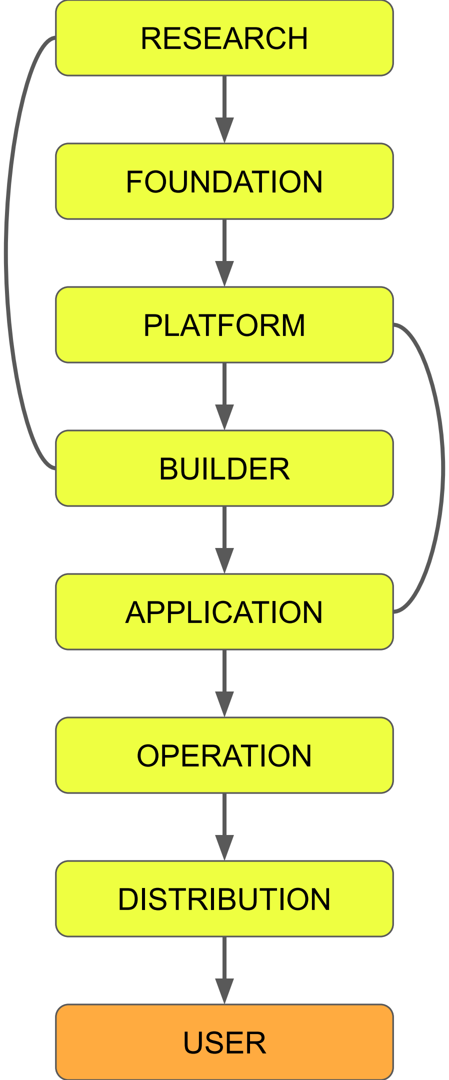
The user layer is where users interact with AI

Purpose

To create value and gather insights & feedback that guide the continuous improvement of AI.

Who Works Here

Daily users, professional users, Domain specific users



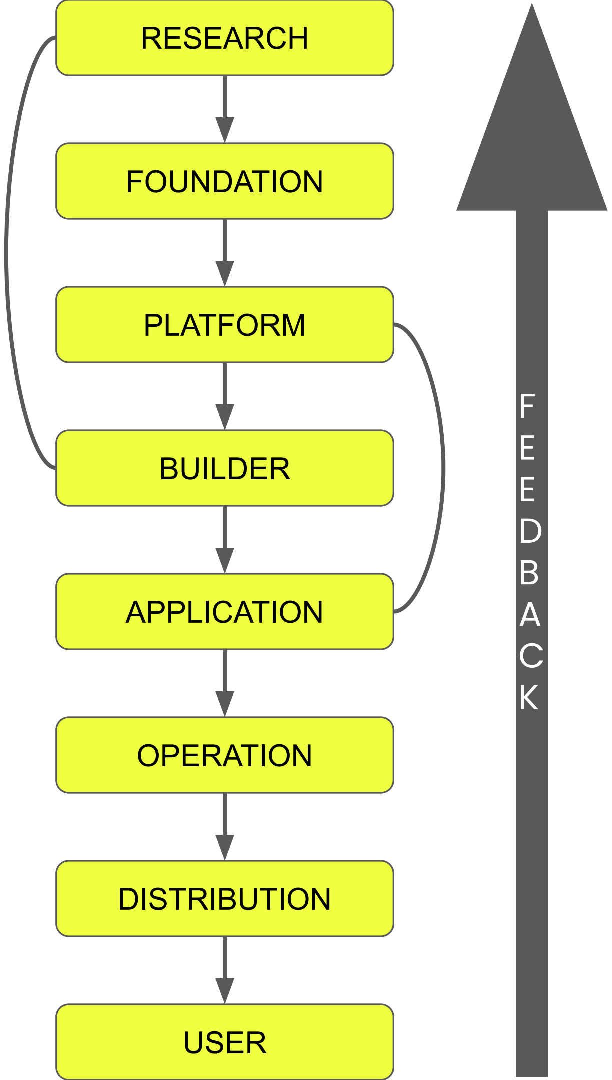
User Layer

Focus Areas

- User Interaction
- Personalization
- Prompt Literacy
- Trust & Ethics
- Feedback Collection

Output

User data, preferences and behavioral data



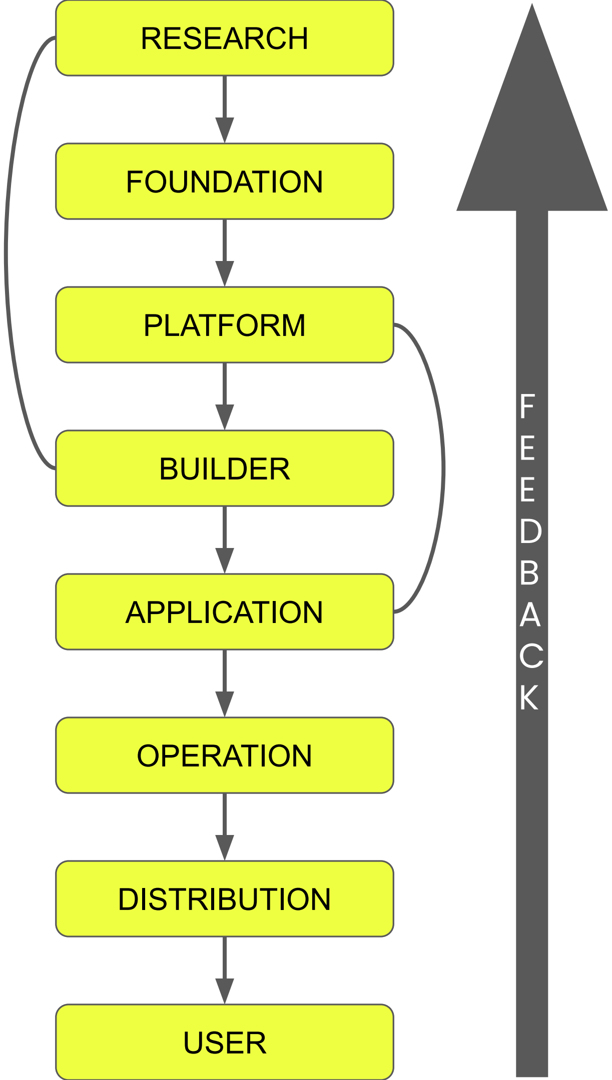
Feedback

User Layer

- Researcher flags fake citations.
- Clicks 👎 and notes "These papers don't exist."
- A small action that triggers a massive feedback chain.

Operation Layer

- Thousands of similar reports detected via dashboards.
- Normal Latency and token usage → this isn't a compute error.
- Identified as a **systemic factuality problem** and gets escalated.



Feedback

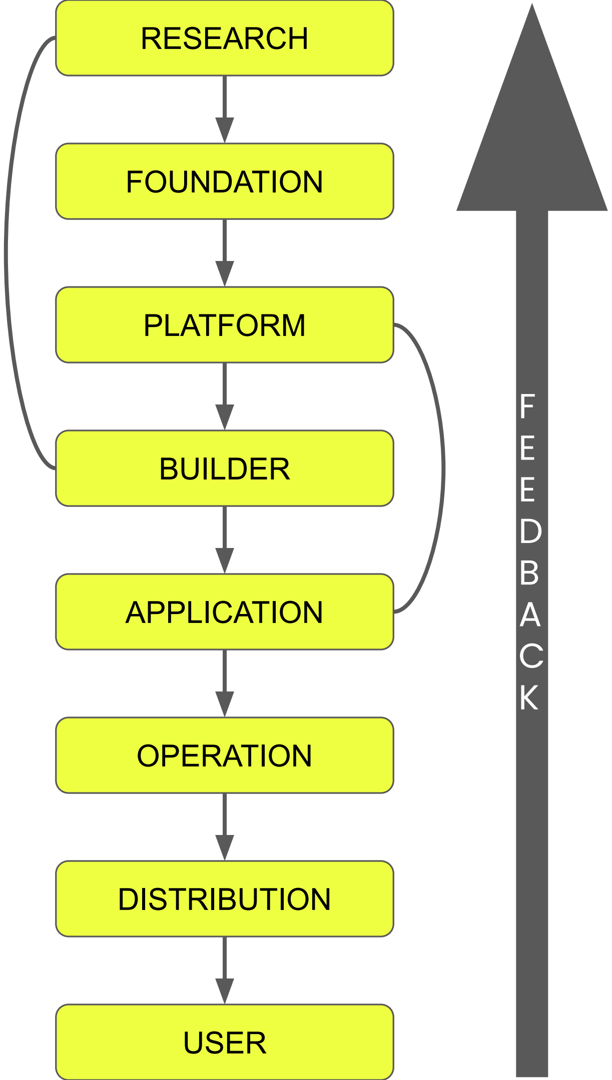
Application Layer

- Prompt templates updated
- Product team adds “Verify sources” disclaimer.
- Limited improvement → escalated to Builder Layer.

Builder Layer

- Teams inspect RAG pipelines (LangChain, LlamaIndex).

  Retriever works fine; model struggles with multiple contexts.
- Issue: reasoning with external data → logged for Platform Layer.



Feedback

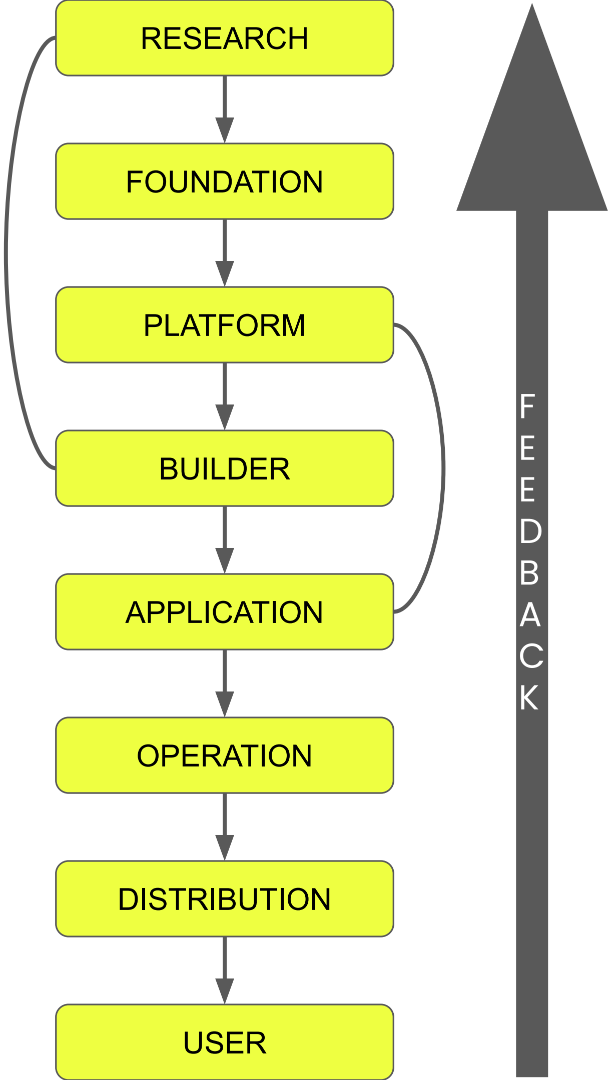
Platform Layer

- API-wide issue confirmed across developer ecosystem.
- High hallucination rate
- Engineers request “factual mode” → escalated to Foundation Layer

Foundation Layer

- Root cause: model training rewarded fluency, not truth.

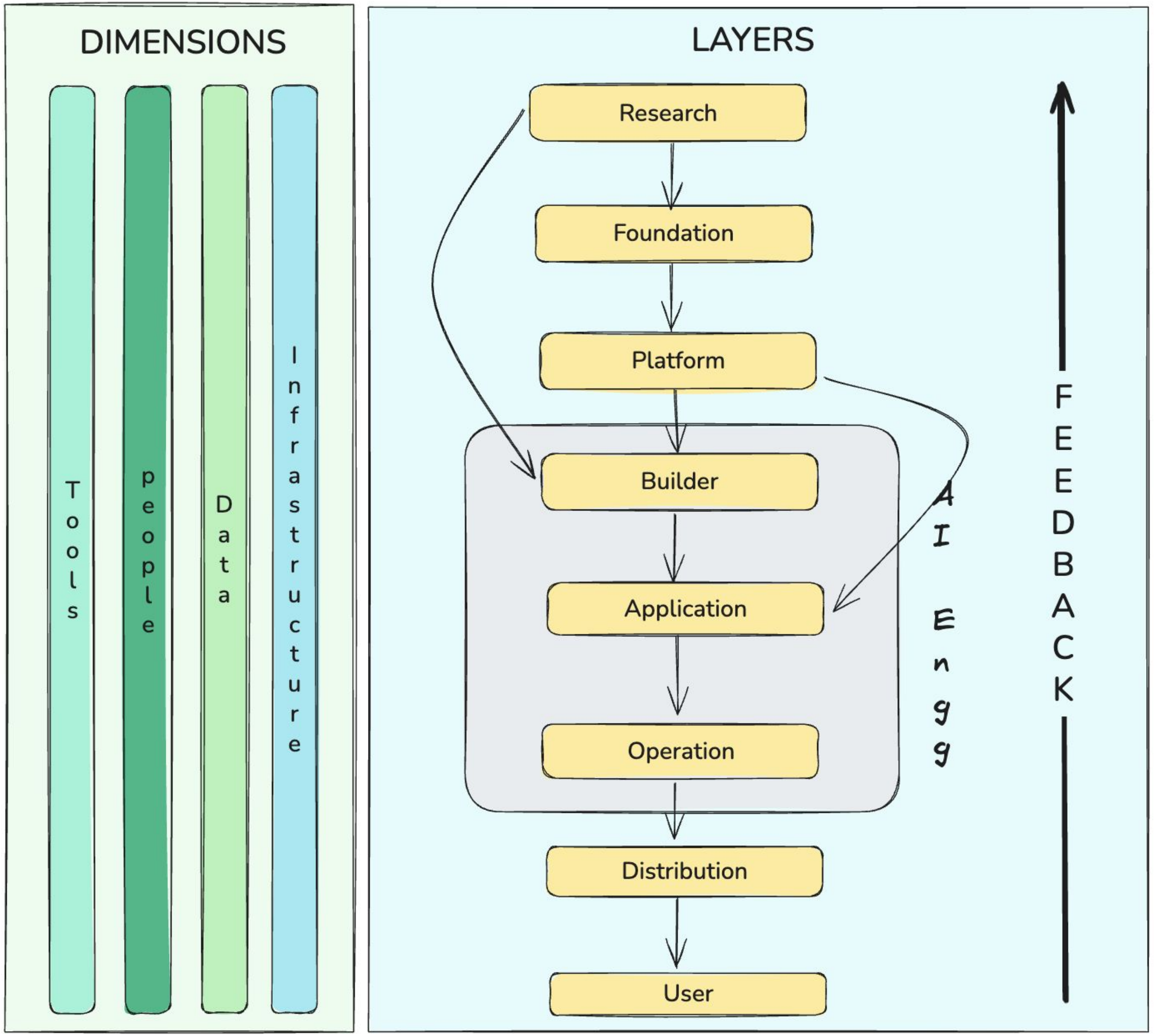
  Fix: retrain with factual data + truthfulness signals.
- Outcome: **GPT-4-Turbo** — fewer hallucinations.

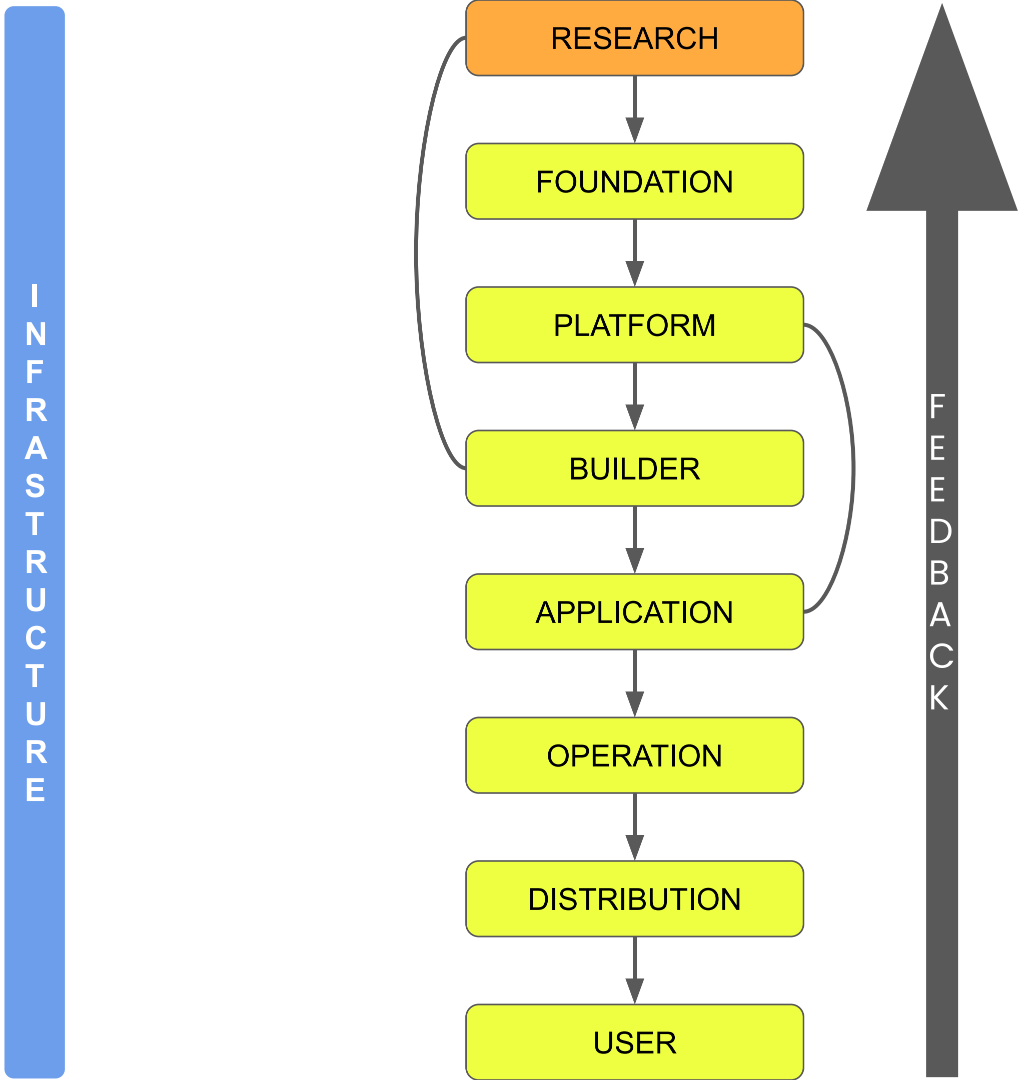


Feedback

Research Layer

- Discovery: models predict **fluency, not truth**.
- New research directions: DPO, Retrieval grounded architecture
- Insights flow into GPT-5, Claude 3 Opus, Gemini 1.5 Pro.

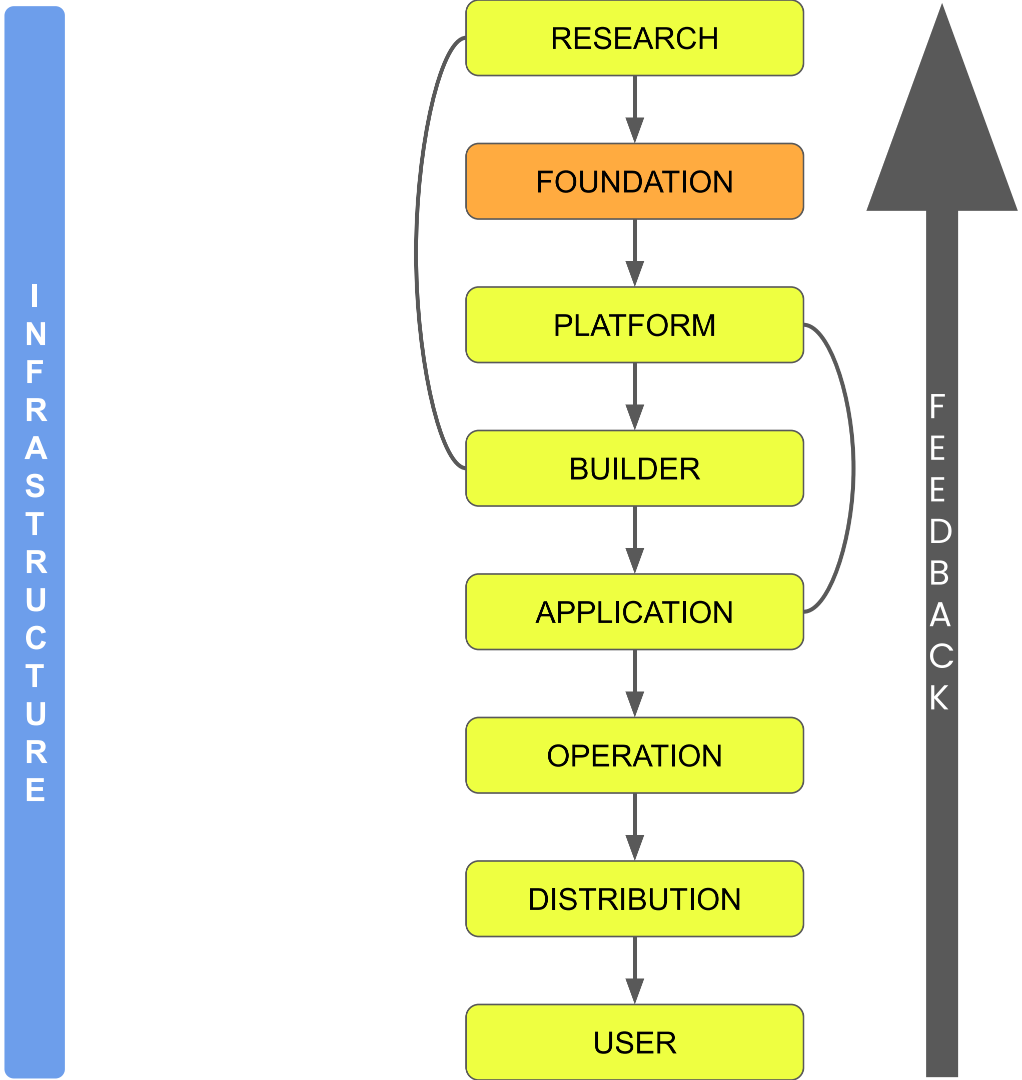

DIMENSIONS
Tools
people
Data
Infrastructure
LAYERS
Research
Foundation
Platform
Builder
Application
Operation
Distribution
User
AI Engg
FEEDBACK



Infrastructure

Research Layer

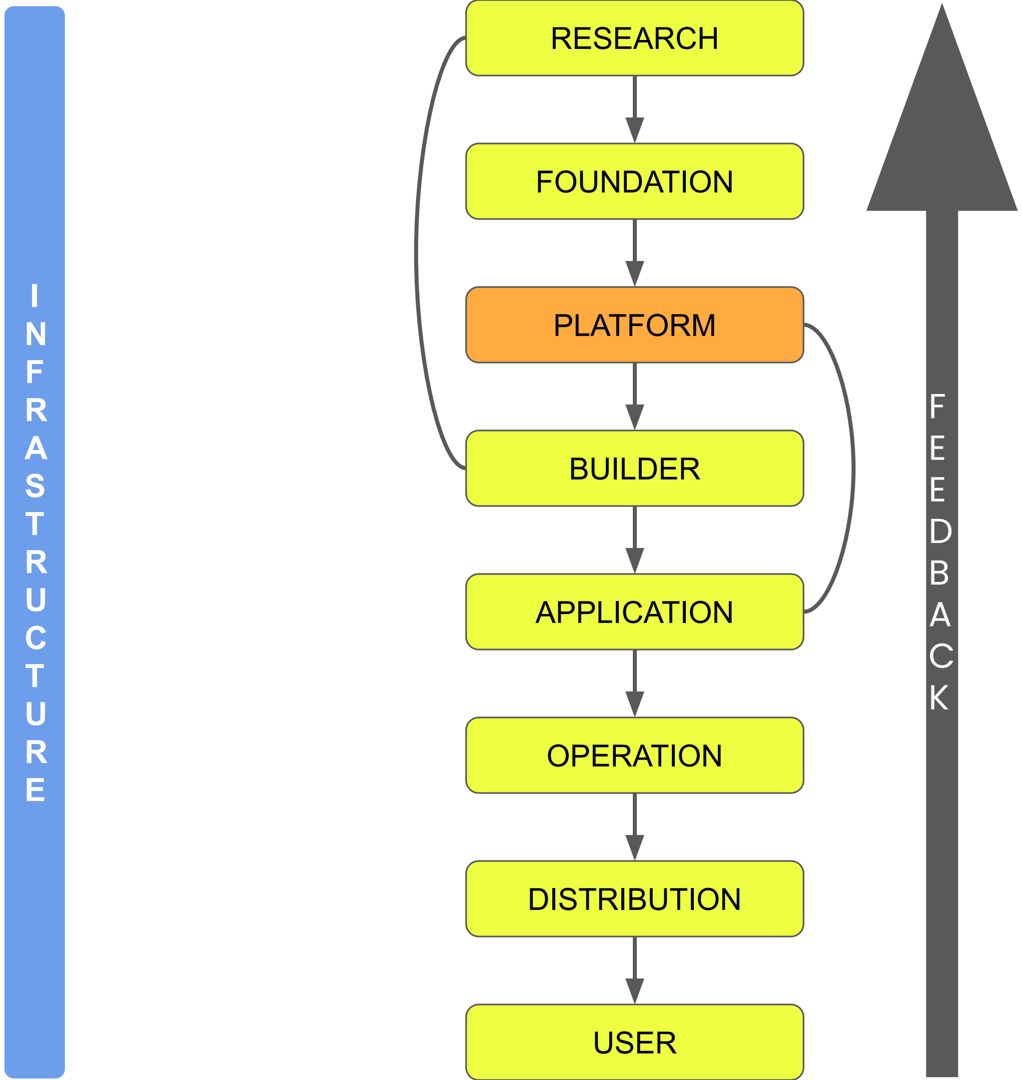
- **Local GPU/TPU Servers** – small scale compute for prototyping.
- **Cloud Compute Platforms** – Resources from **AWS EC2**, **Azure ML**
- **Data Storage** – **Google Cloud Storage**, **S3 buckets** for data, model
- **Version-Controlled Storage** – GitHub, GitLab for code repos.



Infrastructure

Foundation Layer

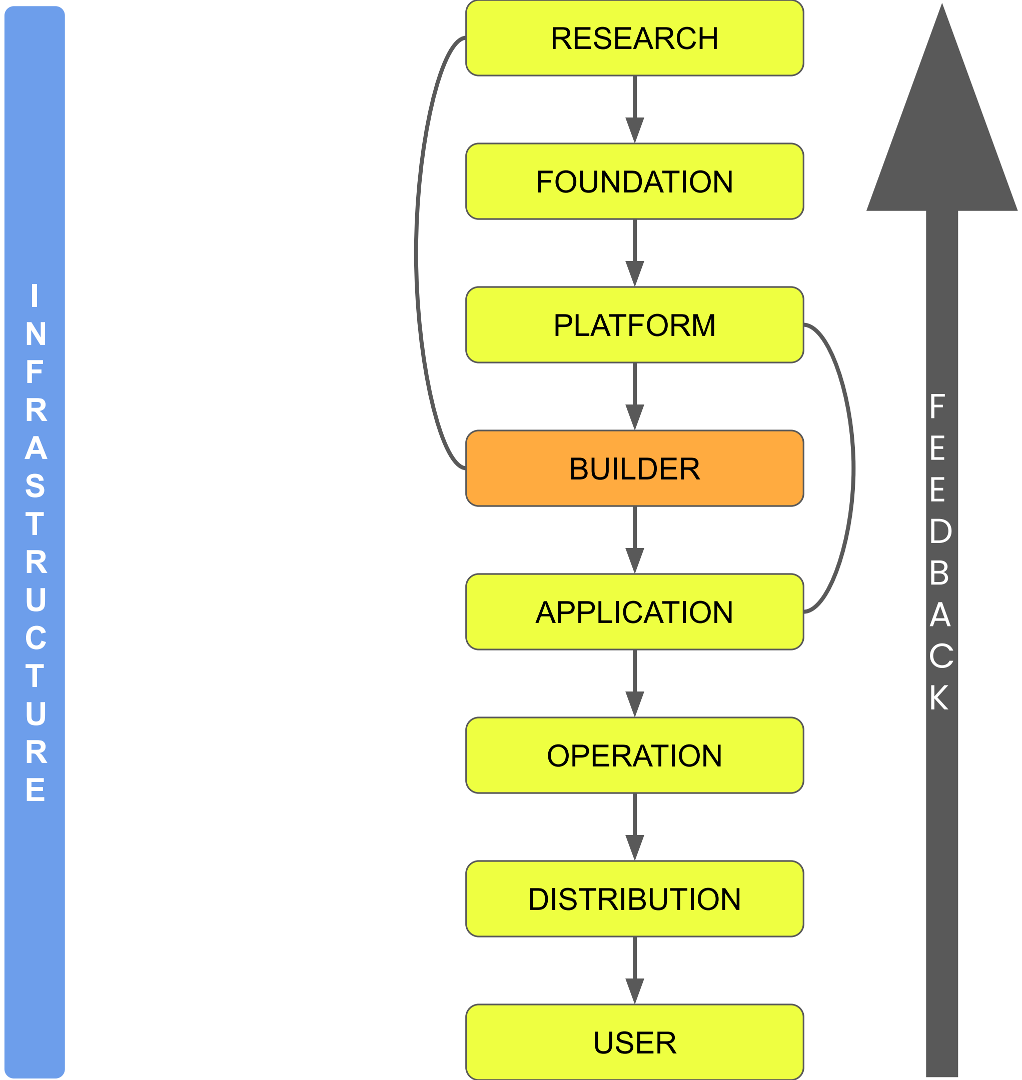
- **GPU/TPU Superclusters** – specialized training pods
- **High-Speed Interconnects** – **InfiniBand**, **NVLink** for parallel data throughput.
- **Distributed File Systems** – **Ceph**, **BeeGFS** for high-volume model training data.
- **Orchestration Systems** – **Kubernetes**, **Ray** for scheduling massive training jobs



Infrastructure

Platform Layer

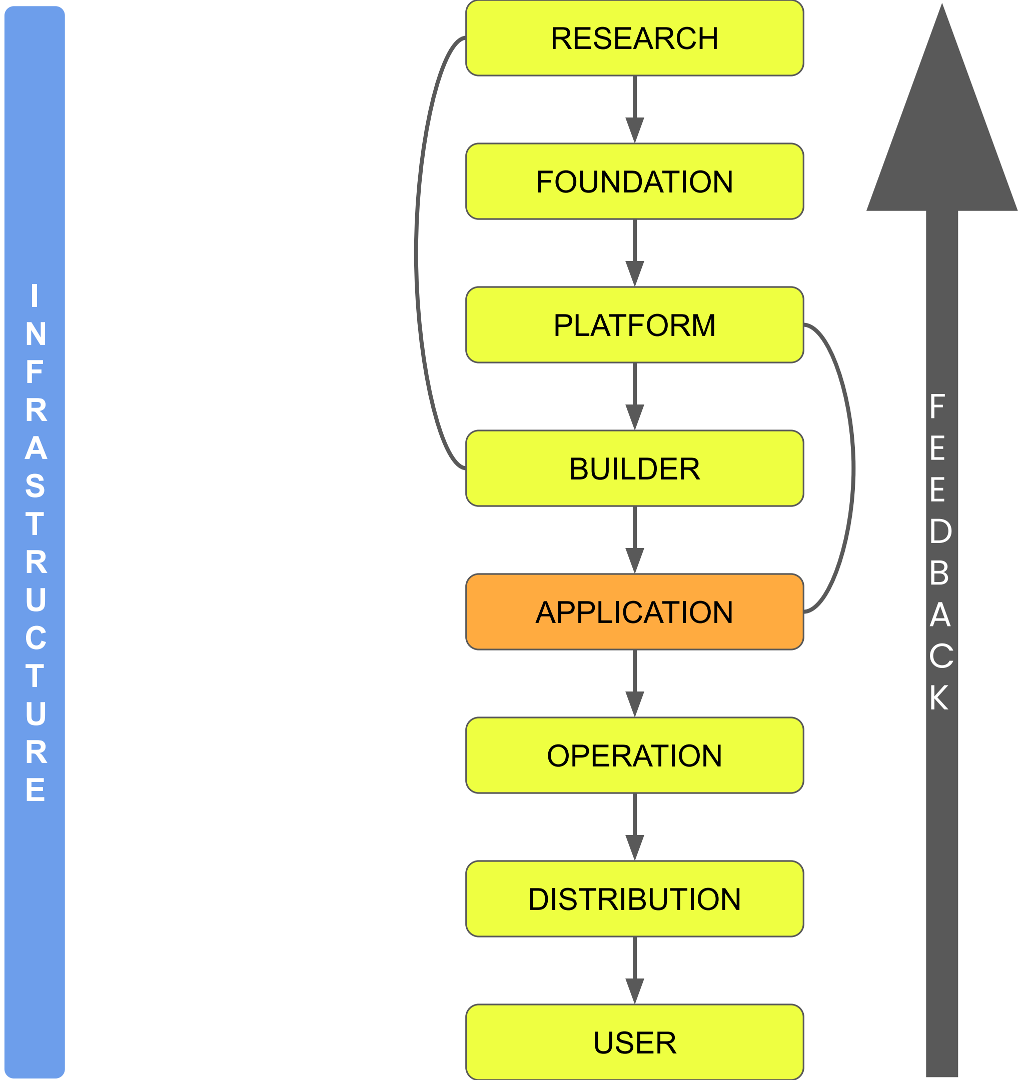
- **Inference Clusters** – GPU/CPU nodes for low-latency serving (e.g., **vLLM**, **TensorRT-LLM**)
- **Model Hosting Infrastructure** – **HF Hub**, **Ollama**, or in-house registries for model artifacts.
- **Load Balancers & CDNs** – **Cloudflare**, **AWS CloudFront** to handle global traffic.
- **Secure Networking** – **VPCs**, firewalls, and VPNs for secure API communication.



Infrastructure

Builder Layer

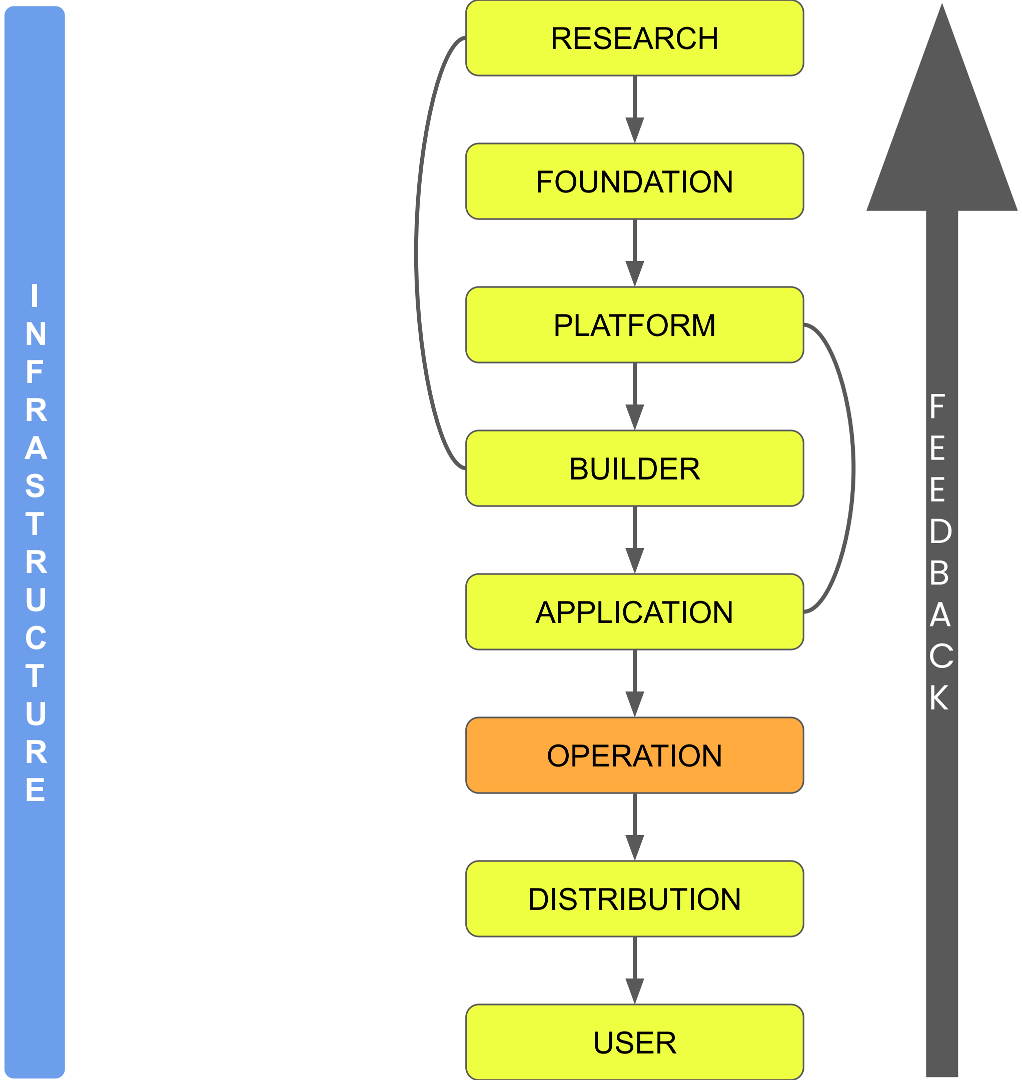
- **Application Servers** – lightweight hosting on **FastAPI**, **Flask**, or **Node.js** for orchestration services.
- **Vector Databases** – **Pinecone**, **Weaviate**, **Chroma**, **FAISS** for RAG.
- **Containerized Builders** – **Docker Compose**, **Kubernetes Pods**, or **Render/Vercel** instances for workflow hosting.



Infrastructure

Application Layer

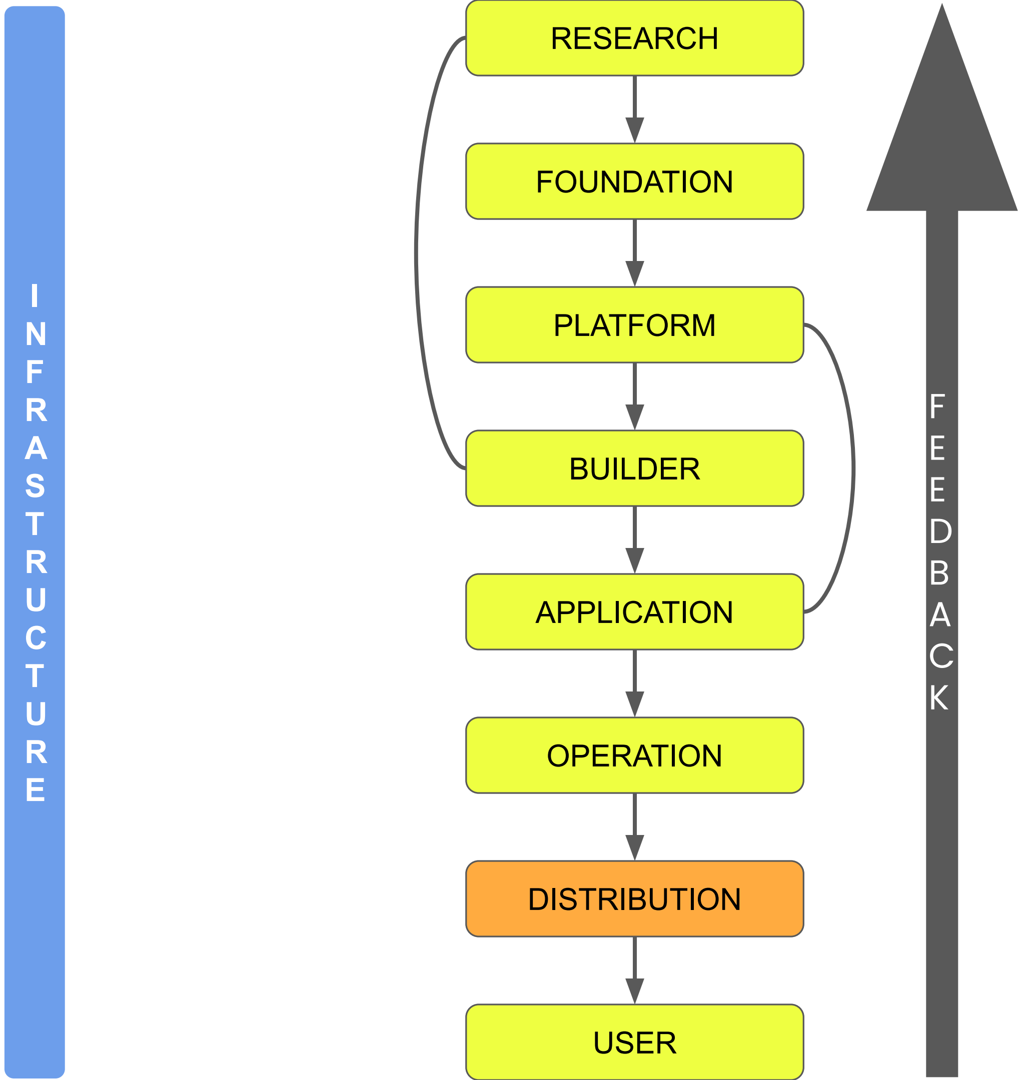
- **Database & Cache Infras– Firebase**, **Redis**, **PostgreSQL** for app state management.
- **CI/CD Pipelines** – **GitHub Actions**, for automated testing and deploy.
- **User Authentication Infrastructure** – **OAuth**, **Firebase Auth**, **Okta** for secure access control
- **Version-Controlled Storage** – GitHub, GitLab for code repos.



Infrastructure

Operation Layer

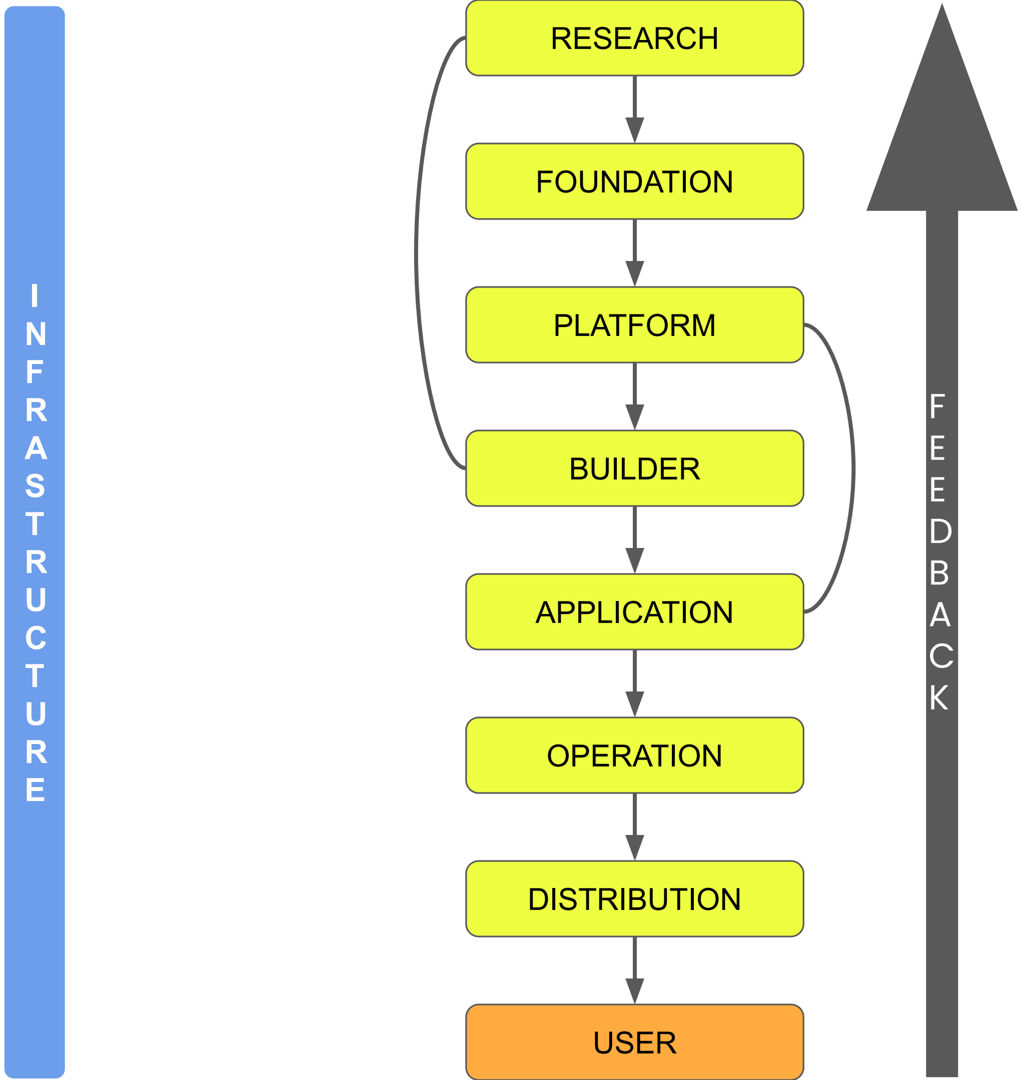
- **Compute:** auto scaling GPU/CPU nodes (containers/K8s).
- **Storage:** object store (artifacts, prompts, logs)
- **Networking:** API gateways, load balancers, private networking/VPC.



Infrastructure

Distribution Layer

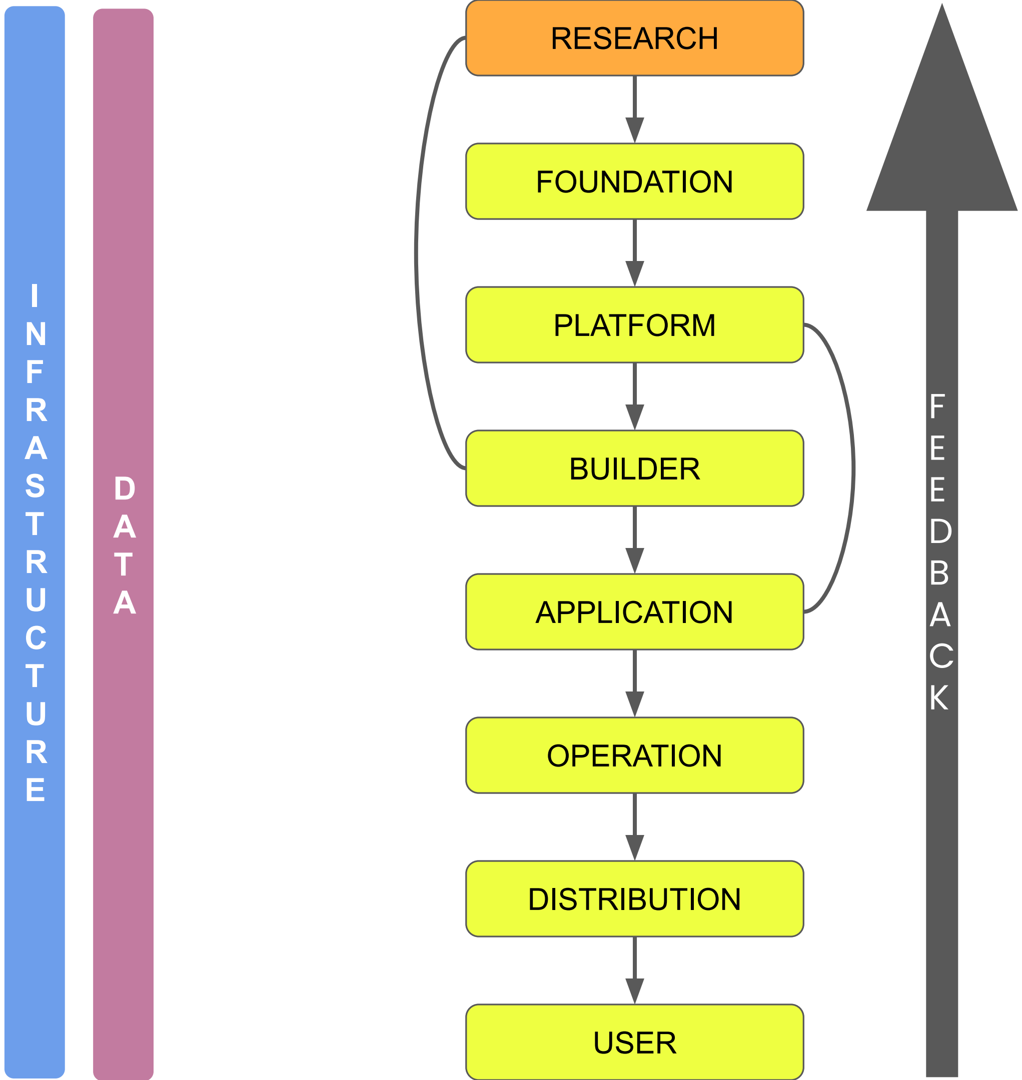
- **App Store Infrastructure** – Apple App Store, Google Play, Slack Marketplace, etc.
- **Billing & Subscription Systems** – **Stripe** for API monetization.
- **Analytics Infrastructure** – **Mixpanel**, **Amplitude**, **Segment** for usage tracking.



Infrastructure

User Layer

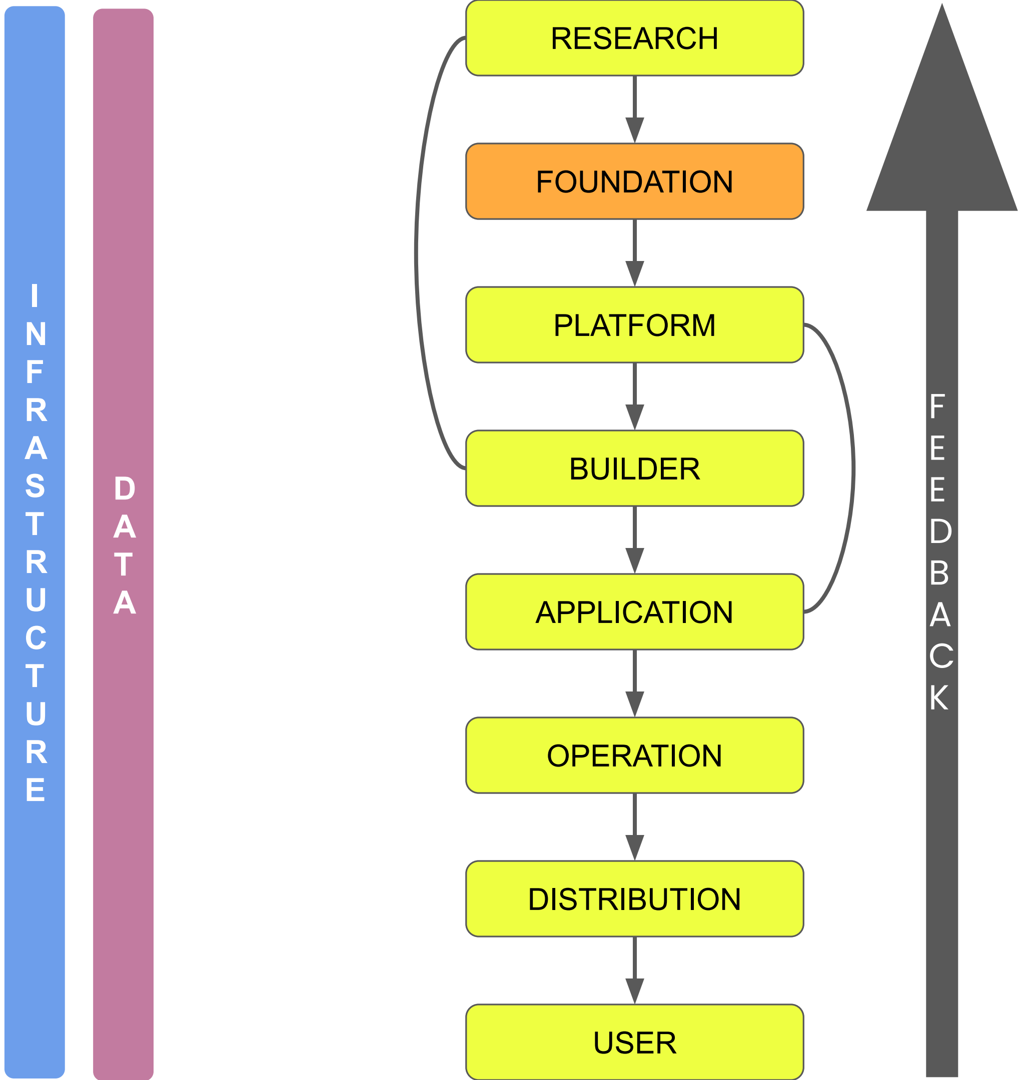
- **End-User Devices** – desktops, smartphones, and edge devices.
- **Network Access** – broadband, 5G, Wi-Fi infrastructure for real-time interaction.
- **Personalization & Storage Infrastructure** – local caches, cookies, and encrypted storage for preferences.



Data

Research Layer

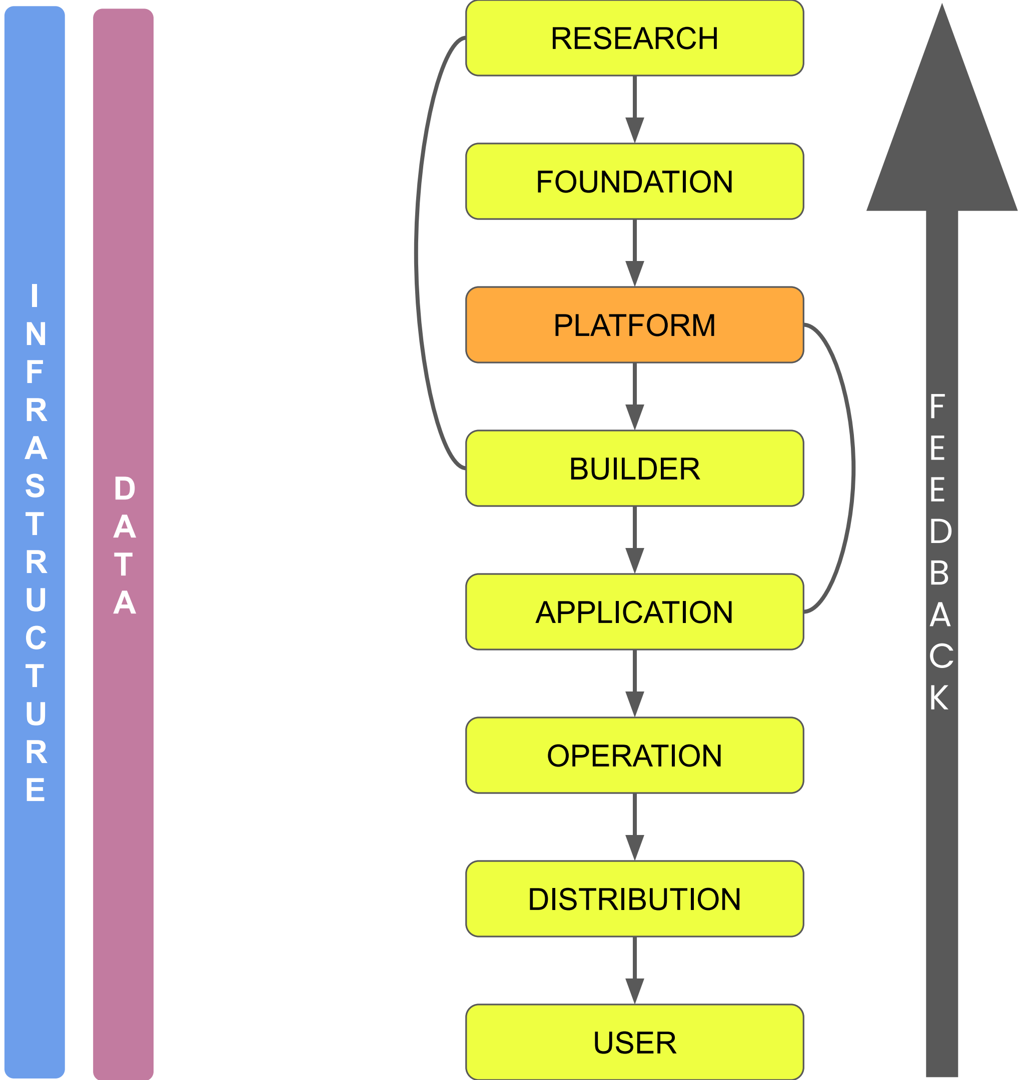
- **Benchmark Datasets** – standardized datasets for model evaluation (e.g., **MNIST**, **CIFAR-10**, **ImageNet**, **GLUE**, **SuperGLUE**).
- **Task-specific datasets** -e.g., **SQuAD** for QA, **COCO** for image captioning.
- **Synthetic Data** – generated data for controlled experimentation and model debugging.



Data

Foundation Layer

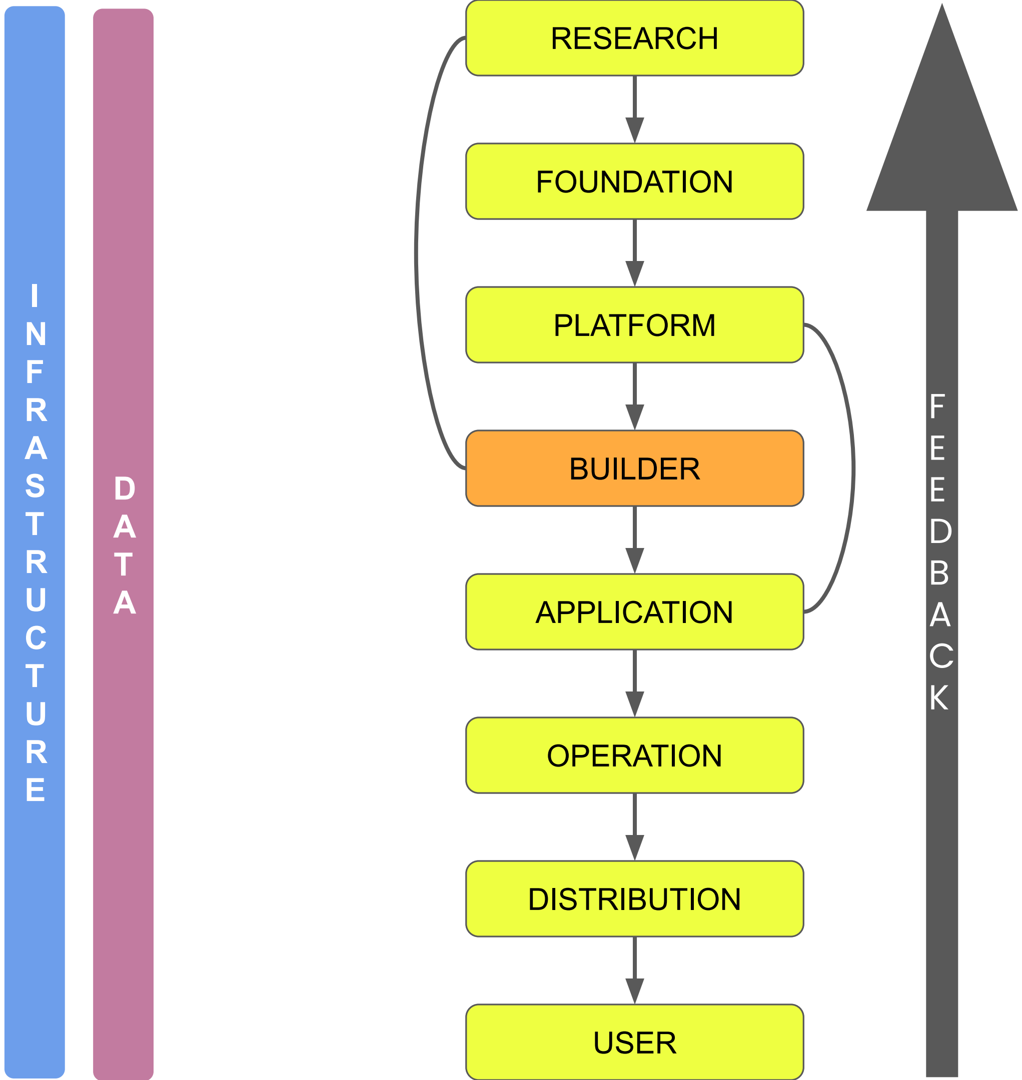
- **Web-Scale Text Corpora** – large-scale crawls (e.g., **Common Crawl**, **Wikipedia**)
- **Multimodal Data** text-image/audio datasets for models like CLIP or Gemini.
- **Alignment Labels** – human preference data (for RLHF), toxicity annotations, and content classification.



Data

Platform Layer

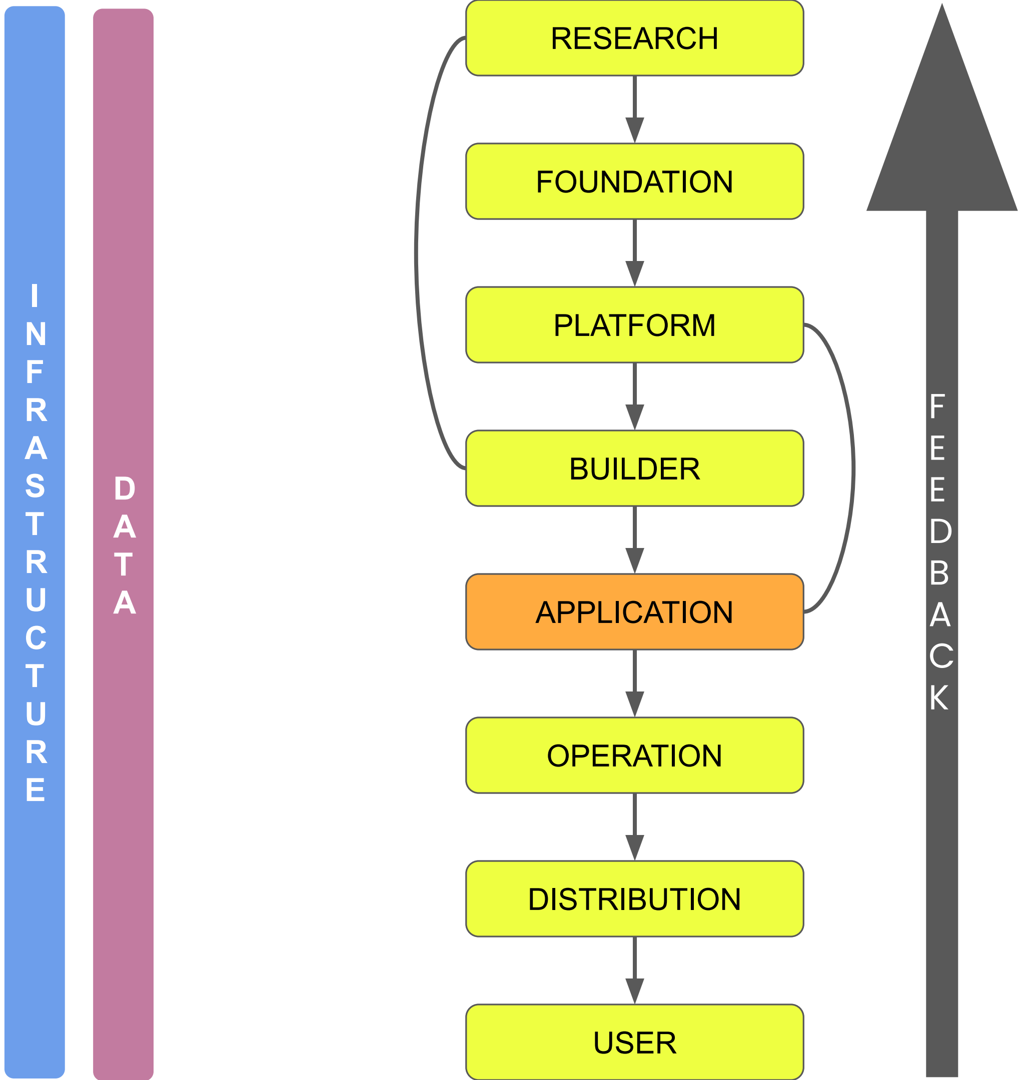
- **Model Artifacts** – serialized model weights, tokenizers, configs.
- **Metrics** – benchmark scores, latency profiles, and cost metrics.
- **Inference Metadata** – logs of queries, responses, latencies, and system performance.



Data

Builder Layer

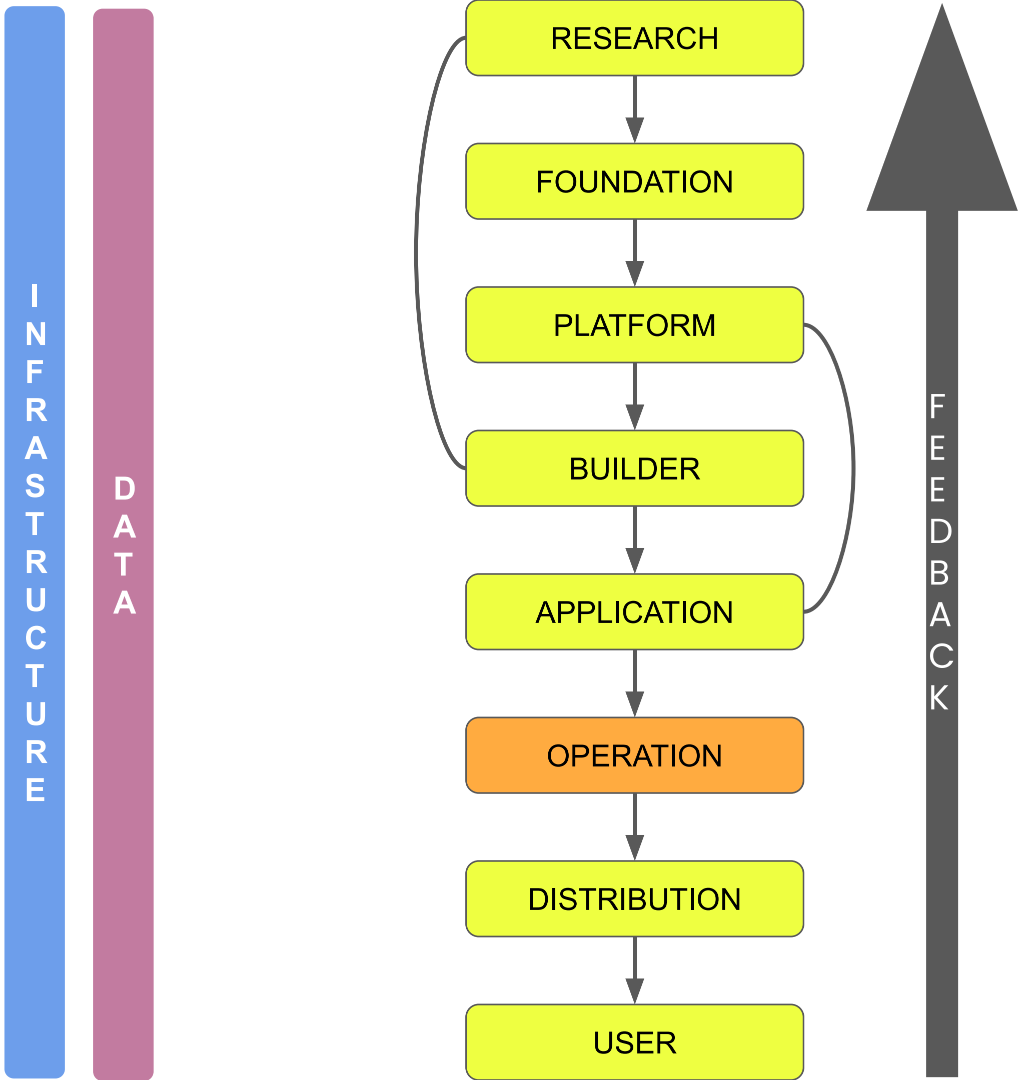
- **Enterprise Knowledge Bases** – wikis, Confluence pages, Notion docs, Slack threads.
- **Vector Databases** – semantic indexes built from enterprise or domain documents (e.g., **Pinecone**).
- **Tool Context Data** – results returned by API calls or function executions.
- **Long-Term Memory Stores** – previous interactions and agent state histories.



Data

Application Layer

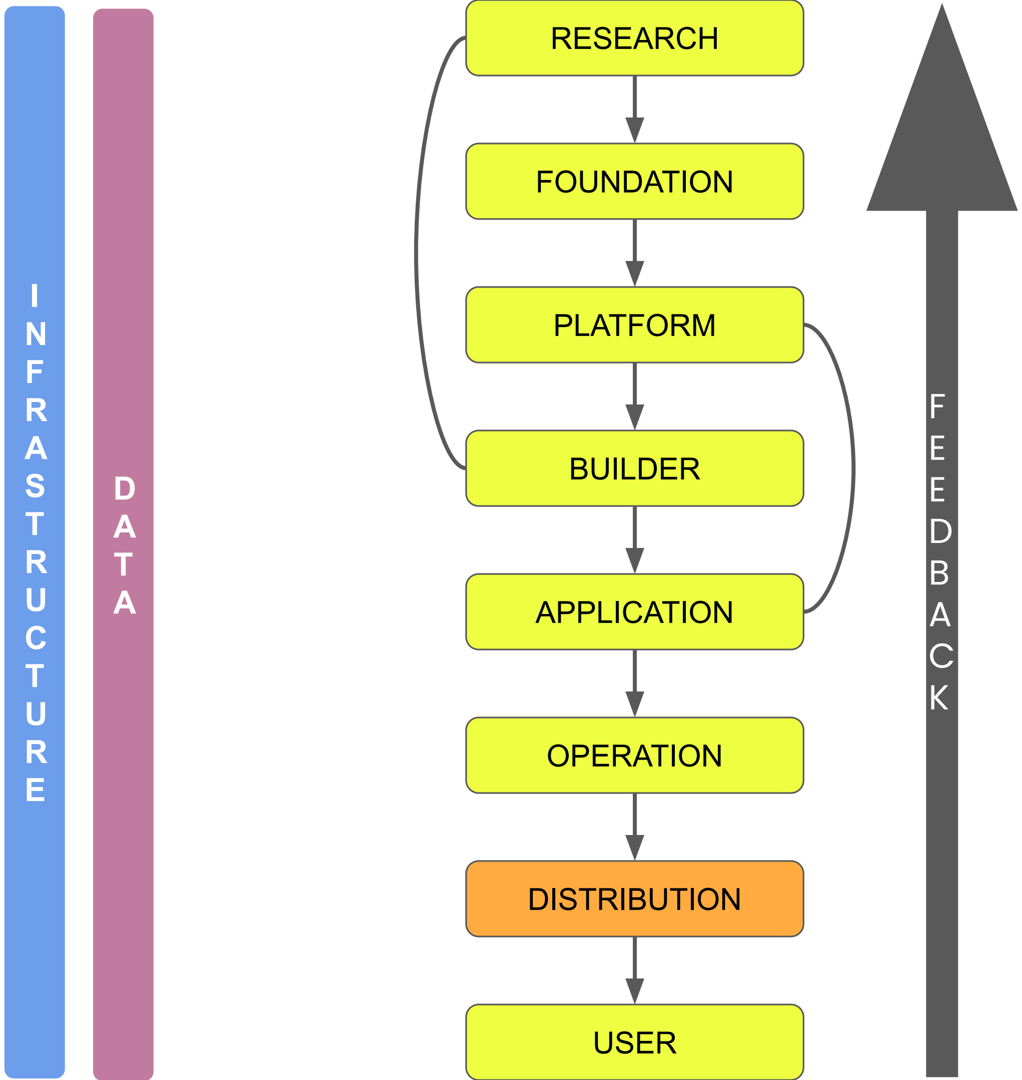
- **User Data** – text prompts, uploaded files, interaction logs.
- **Domain Data** – internal business documents, customer support transcripts, CRM records.
- **Contextual Data Bindings** – organization-specific knowledge linked into app logic.
- **Feedback Data** – user corrections, ratings, and fine-tuning feedback.



Data

Operation Layer

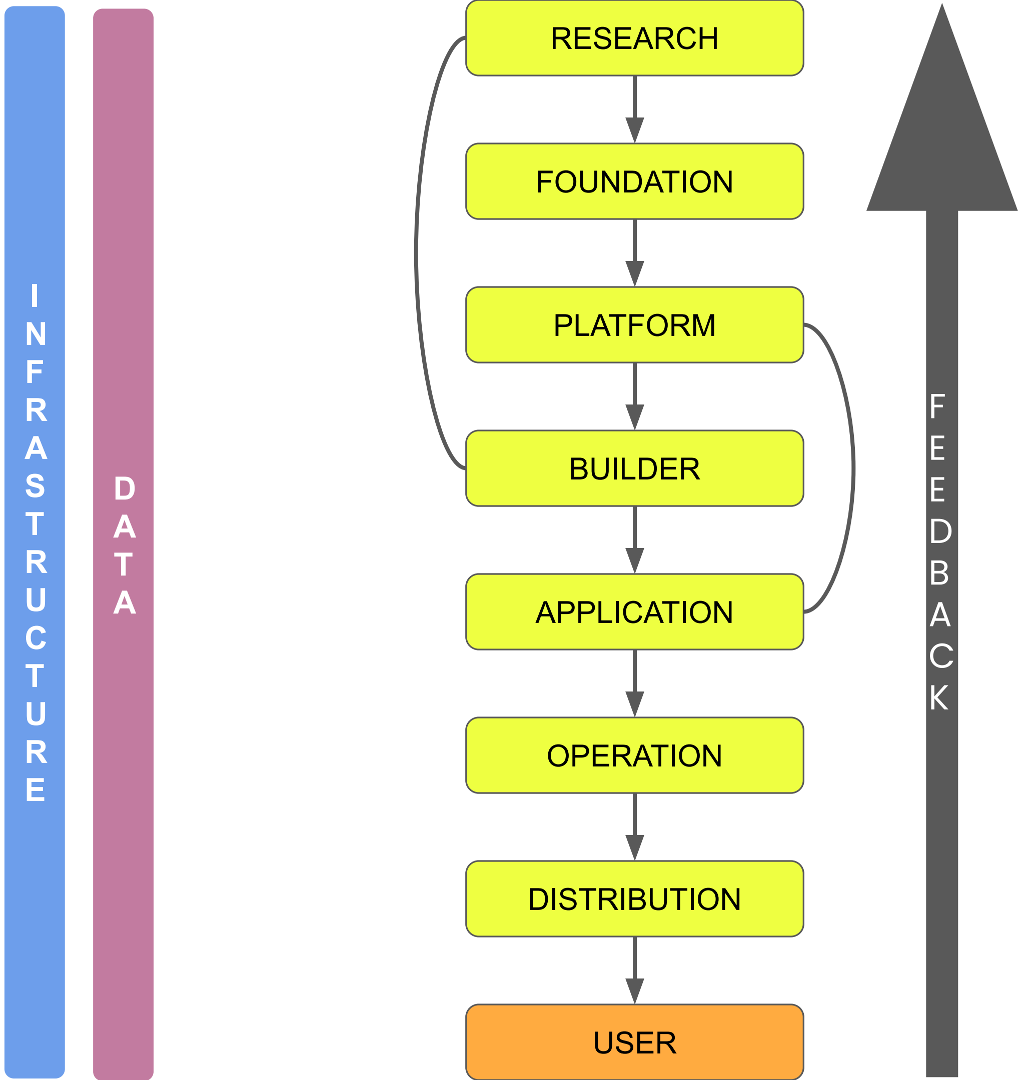
- **Telemetry Streams** – logs from model inference, latency, and resource usage.
- **User Feedback Data** – thumbs-up/down, corrections, and explicit error reports.
- **Evaluation Data** – curated test sets and golden labels for regression testing.



Data

Distribution Layer

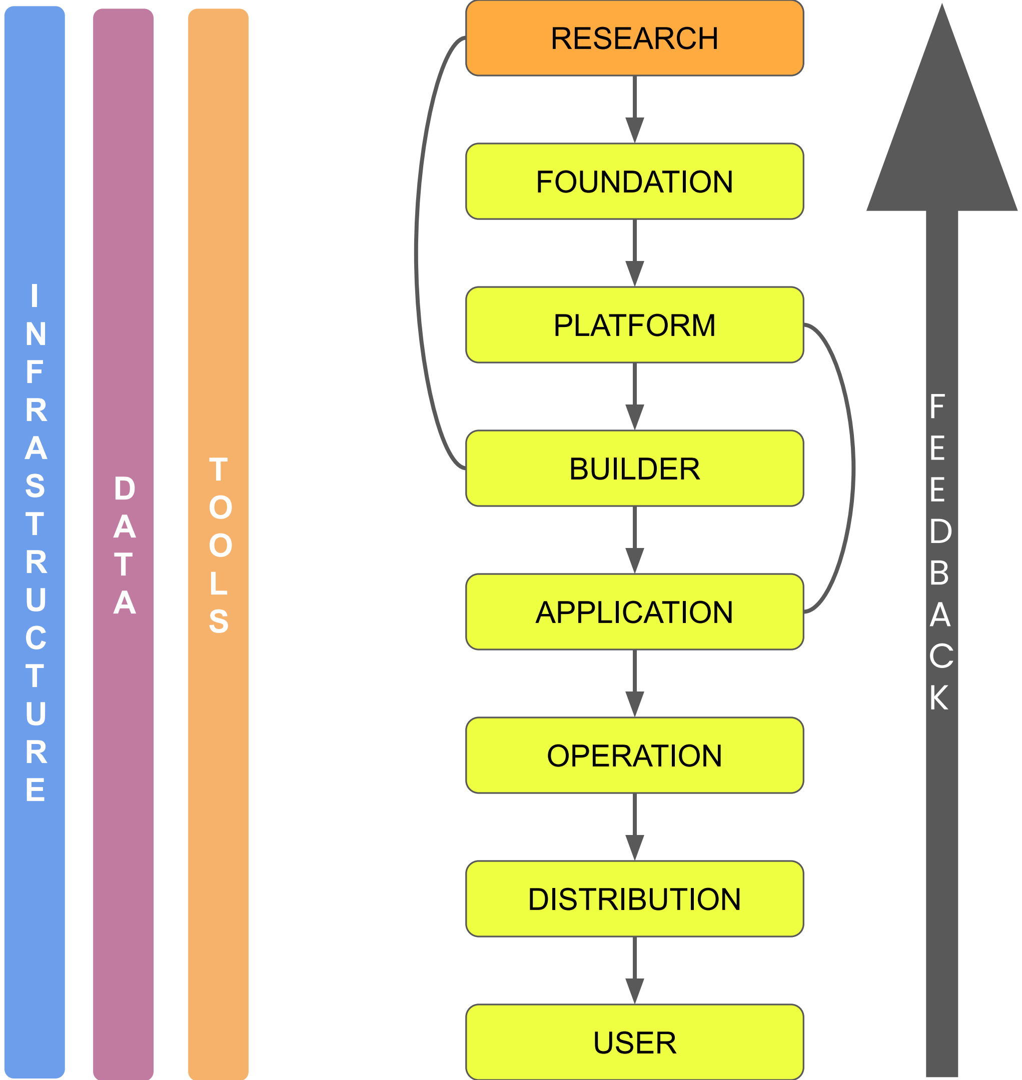
- **User Growth Metrics** – adoption rates, active users, retention curves.
- **Channel Analytics** – app store impressions, API call volume, campaign performance.
- **Revenue Data** – subscription and API billing metrics.
- **A/B Testing Results** – behavioral data comparing different rollout strategies.



Data

User Layer

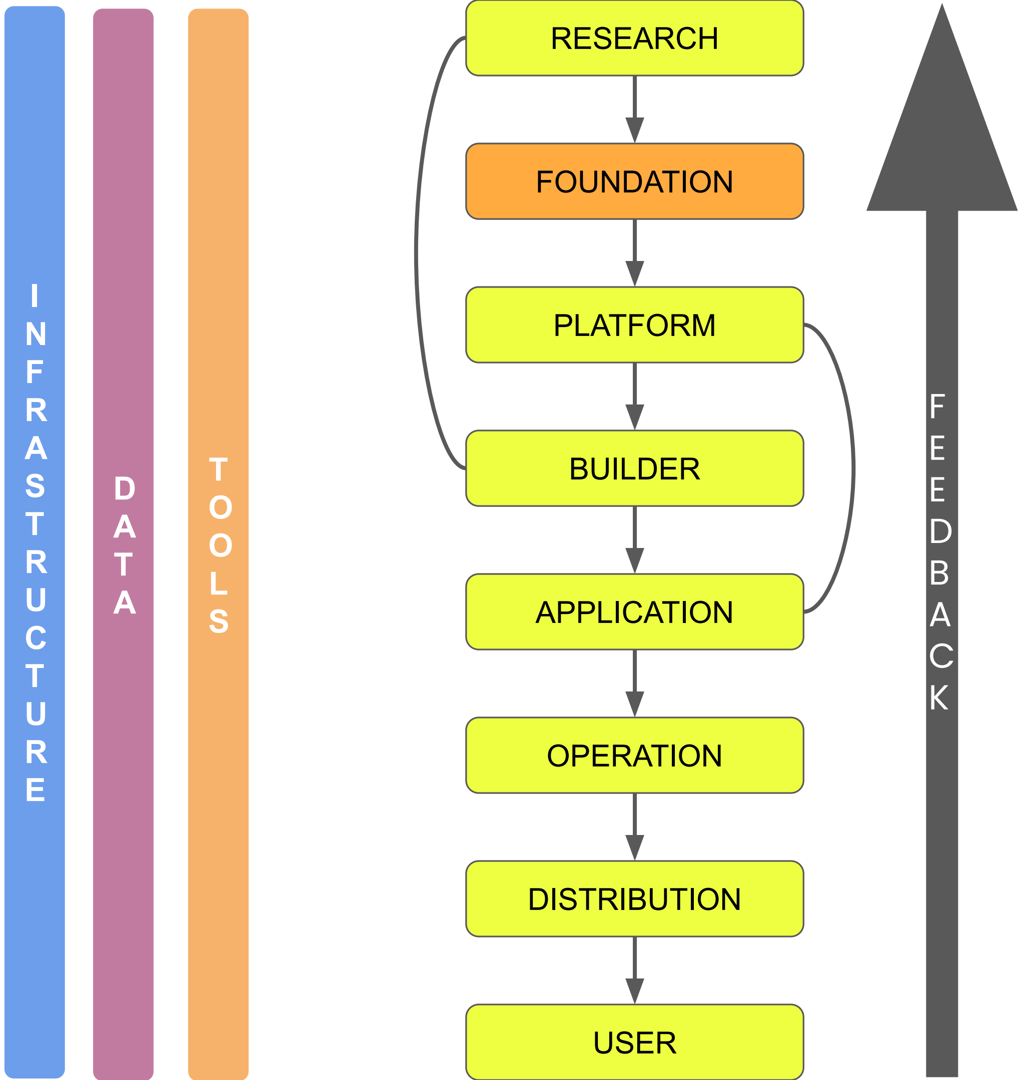
- **User Preferences** – saved configs, domains, or instructions.
- **Behavioral Analytics** – usage frequency, engagement metrics, and satisfaction signals.
- **Feedback & Ratings** – direct user corrections and qualitative comments.
- **Prompt Histories** – previous queries and responses used for contextual continuity.



Tools

Research Layer

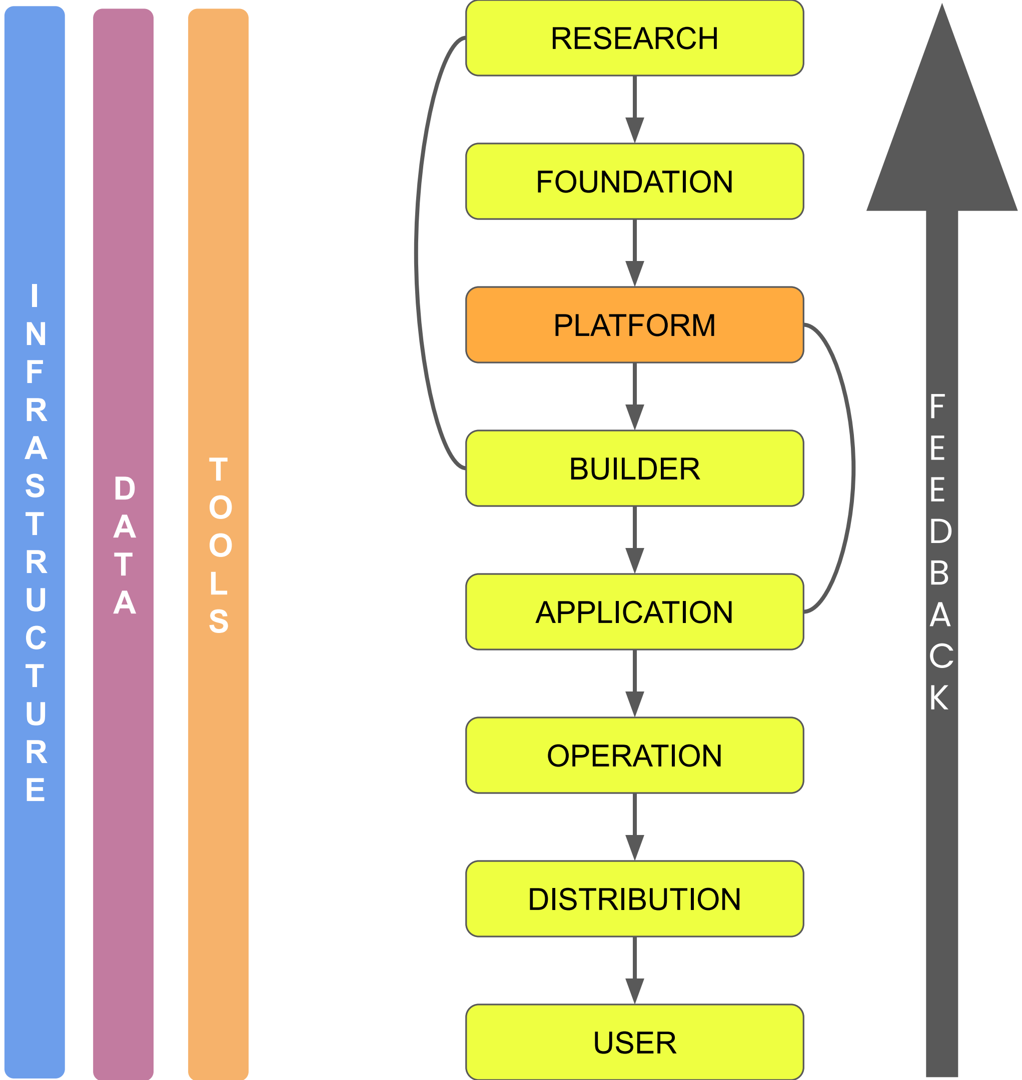
- **Jupyter / Colab** – interactive notebooks for rapid prototyping.
- **PyTorch / TensorFlow** – frameworks for building and experimenting with model architectures.
- **HF Datasets / Papers with Code** – standardized repos of datasets and benchmarks.
- **Weights & Biases (W&B) / MLflow** – experiment tracking and visualization dashboards.



Tools

Foundation Layer

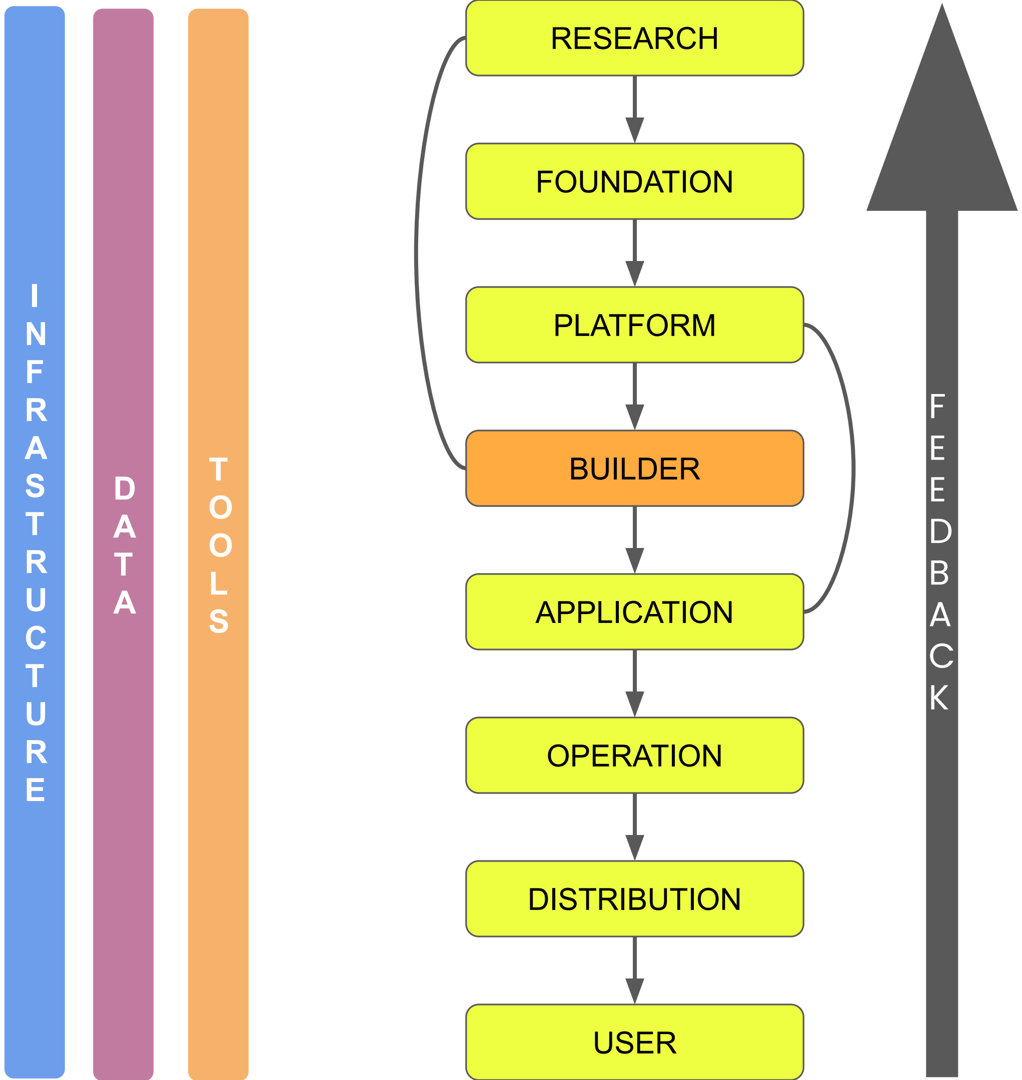
- **PyTorch / TensorFlow** – training engines for large-scale models.
- **DeepSpeed / Megatron-PyTorch Lightning** – distributed training and optimization frameworks.
- **Ray Train / Kubernetes**– tools for managing large GPU clusters.
- **DVC / LakeFS** – data versioning and scalable dataset management.



Tools

Platform Layer

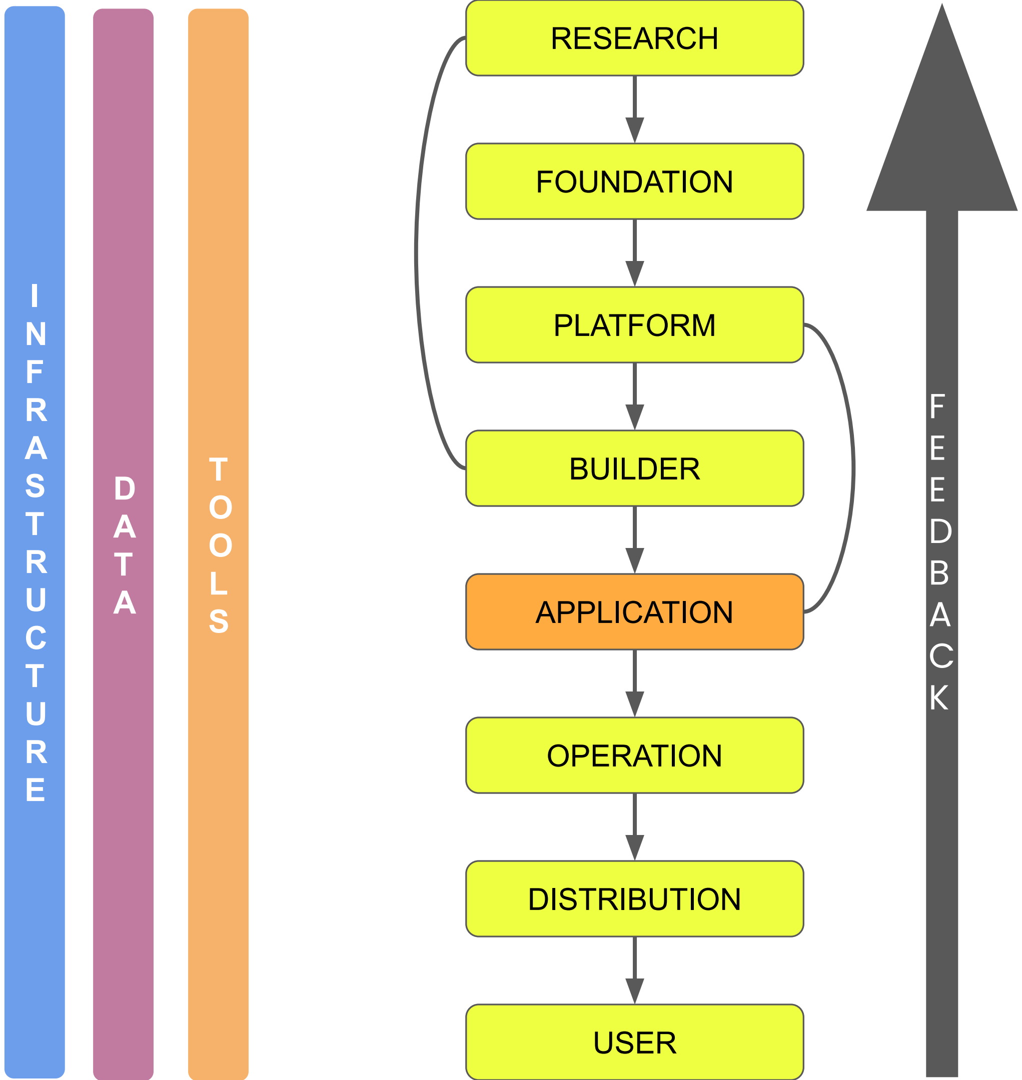
- **vLLM / TensorRT-LLM / ONNX Runtime** – optimized inference engines for large models.
- **Docker / Kubernetes / Helm / Terraform** – containerization and deployment infrastructure.
- **Hugging Face Hub / MLflow Model Registry** – model repositories and registries.
- **FastAPI / gRPC / API Gateway (Apigee)** – frameworks for exposing inference endpoints as APIs.



Tools

Builder Layer

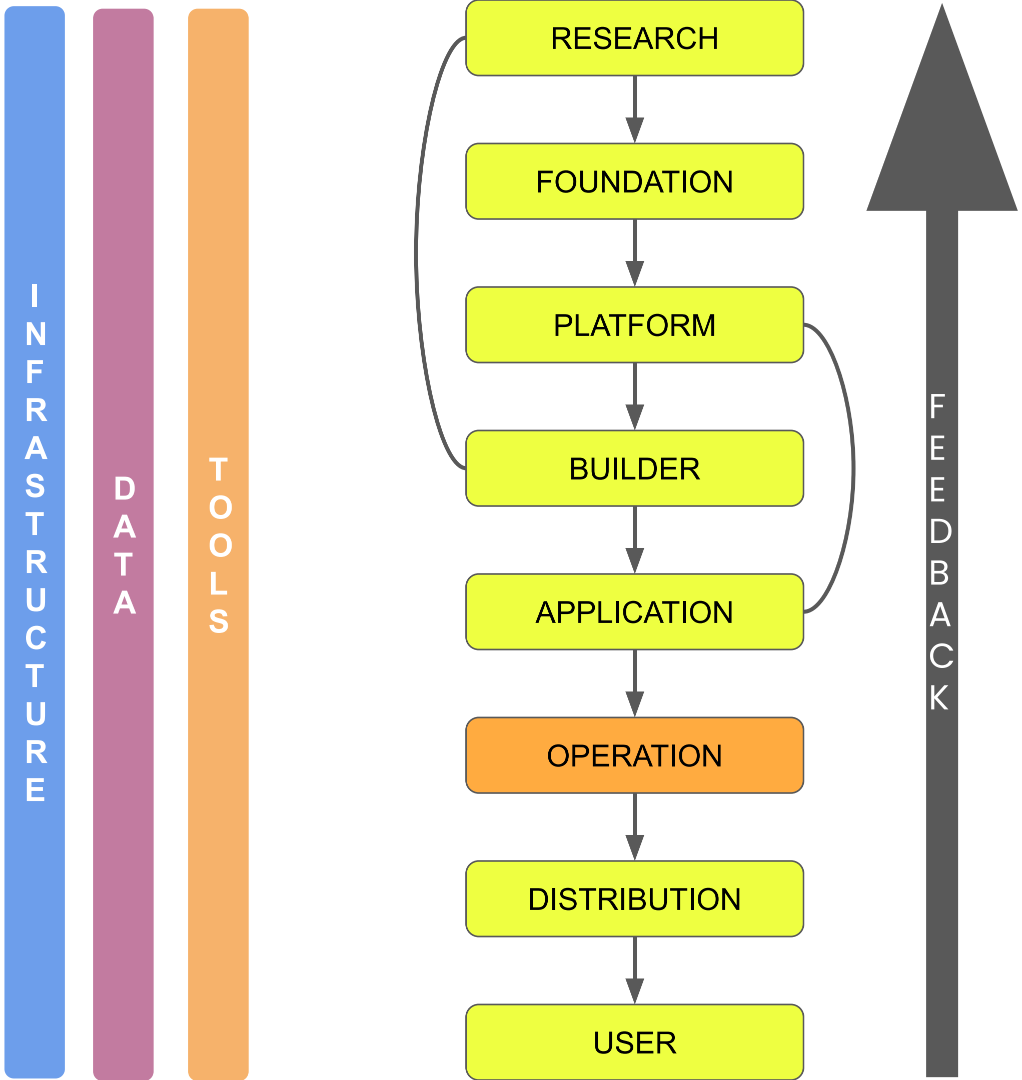
- **OpenAI / Anthropic / Together / Ollama APIs** – foundation model endpoints for inference and embeddings.
- **FastAPI / Flask** – lightweight frameworks to serve orchestrated logic as APIs.
- **Pinecone / Weaviate / FAISS / Chroma** – vector databases for semantic search and retrieval augmentation.



Tools

Application Layer

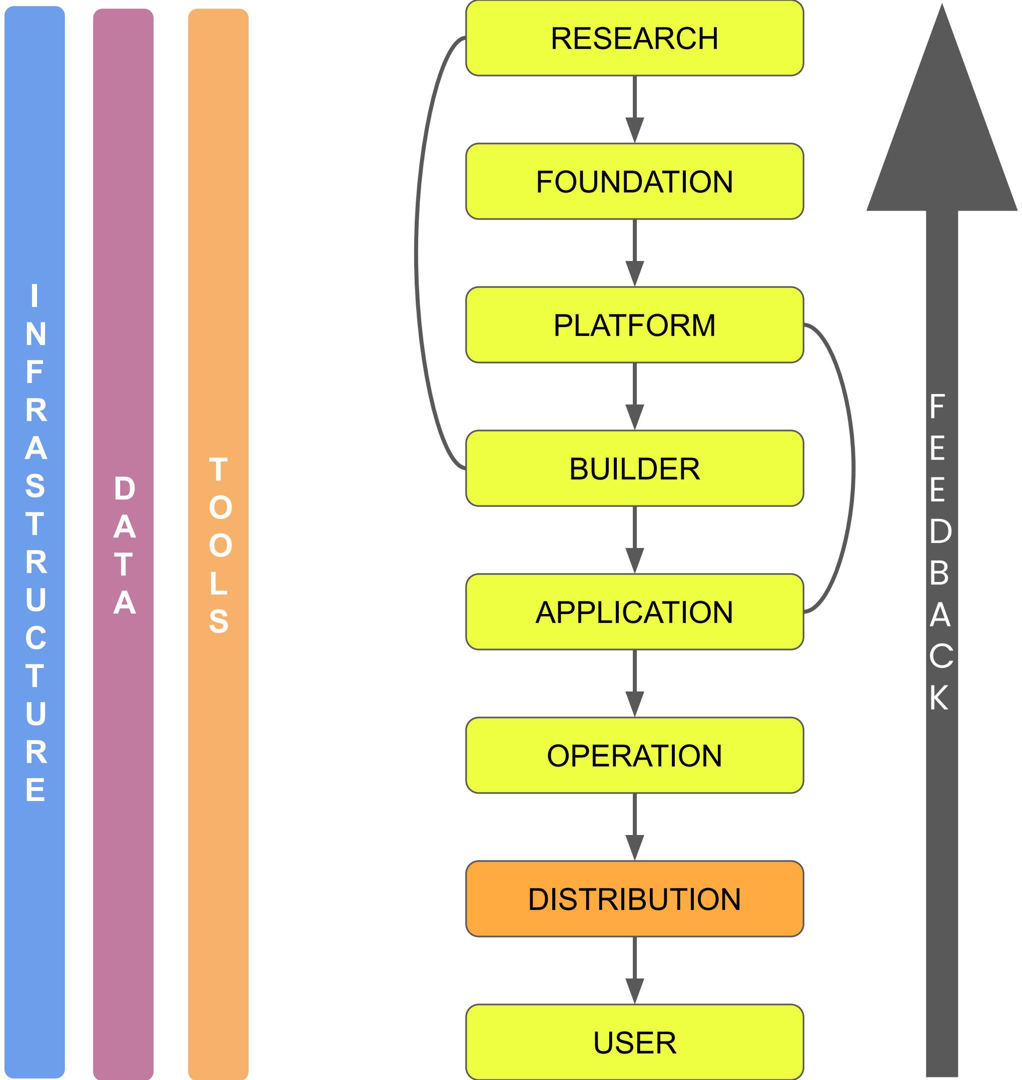
- **LangChain / LangGraph / CrewAI** – orchestration frameworks for multi-step reasoning and RAG.
- **Next.js / React / Flask / FastAPI** – web frameworks for interactive front-ends and APIs.
- **OpenAI API / Anthropic API / Ollama** – model-serving interfaces for embedding AI into products.
- **PromptLayer** – tools for monitoring prompts



Tools

Operation Layer

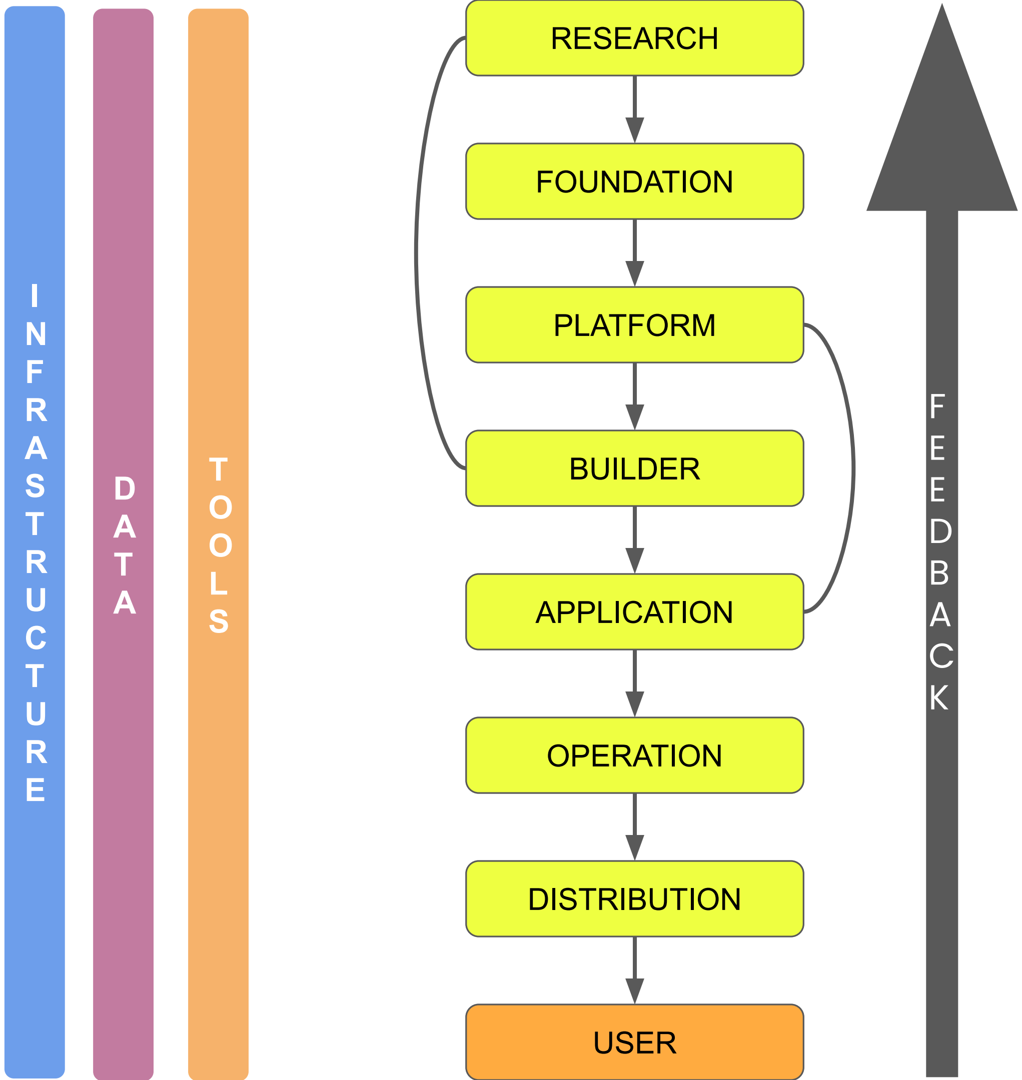
- **Tools for Deployment** – Docker, Kubernetes, Github Actions.
- **LangSmith / LangFuse / Arize AI** – observability and evaluation tools for LLM apps.
- **Prometheus / Grafana** – infrastructure monitoring and performance tracking.
- **Guardrails AI** – enforcing output safety and compliance.



Tools

Distribution Layer

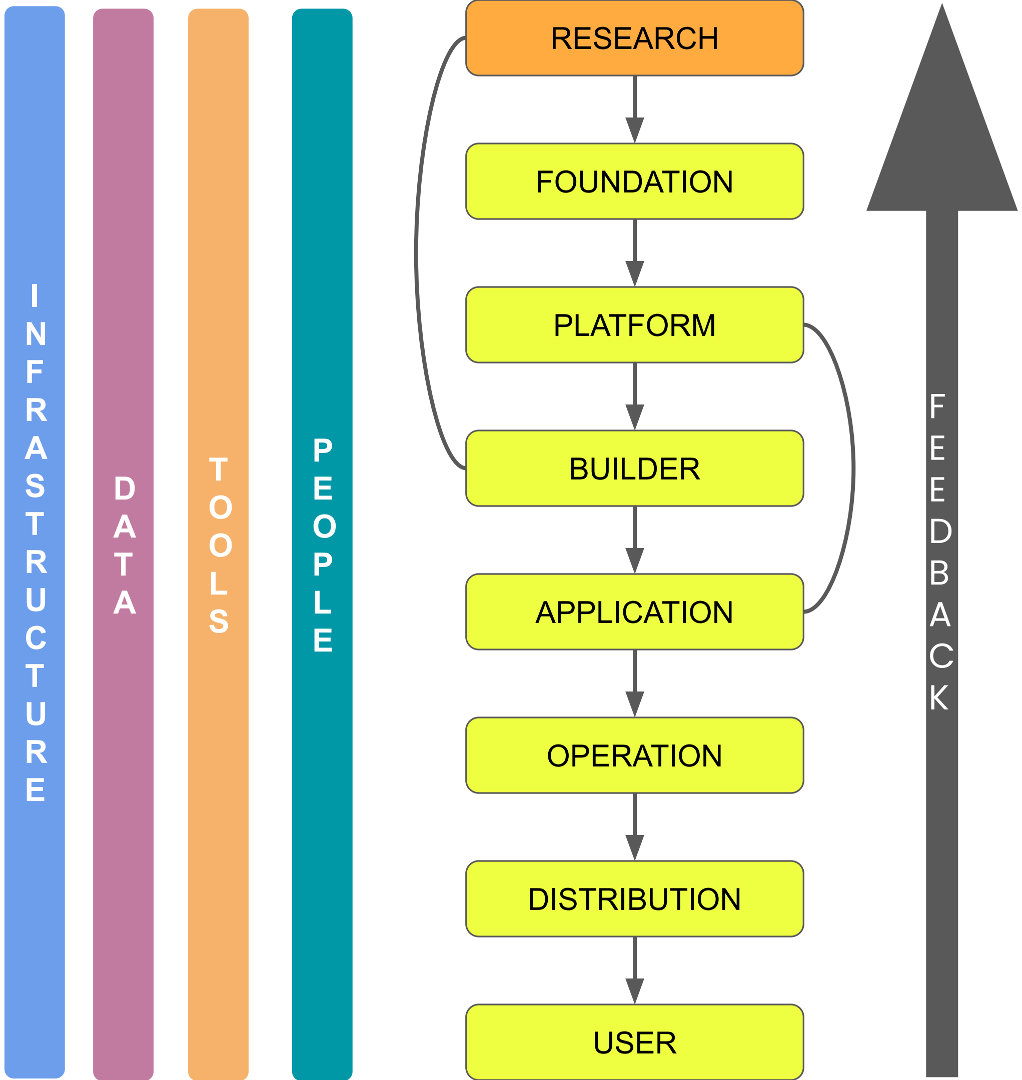
- **App Store Connect / Google Play Console / Slack Marketplace** – distribution platforms for app publishing.
- **API Gateway / CDN** – API exposure, delivery optimization and controlled rollout.
- **Stripe** – For monetization and billing



Tools

User Layer

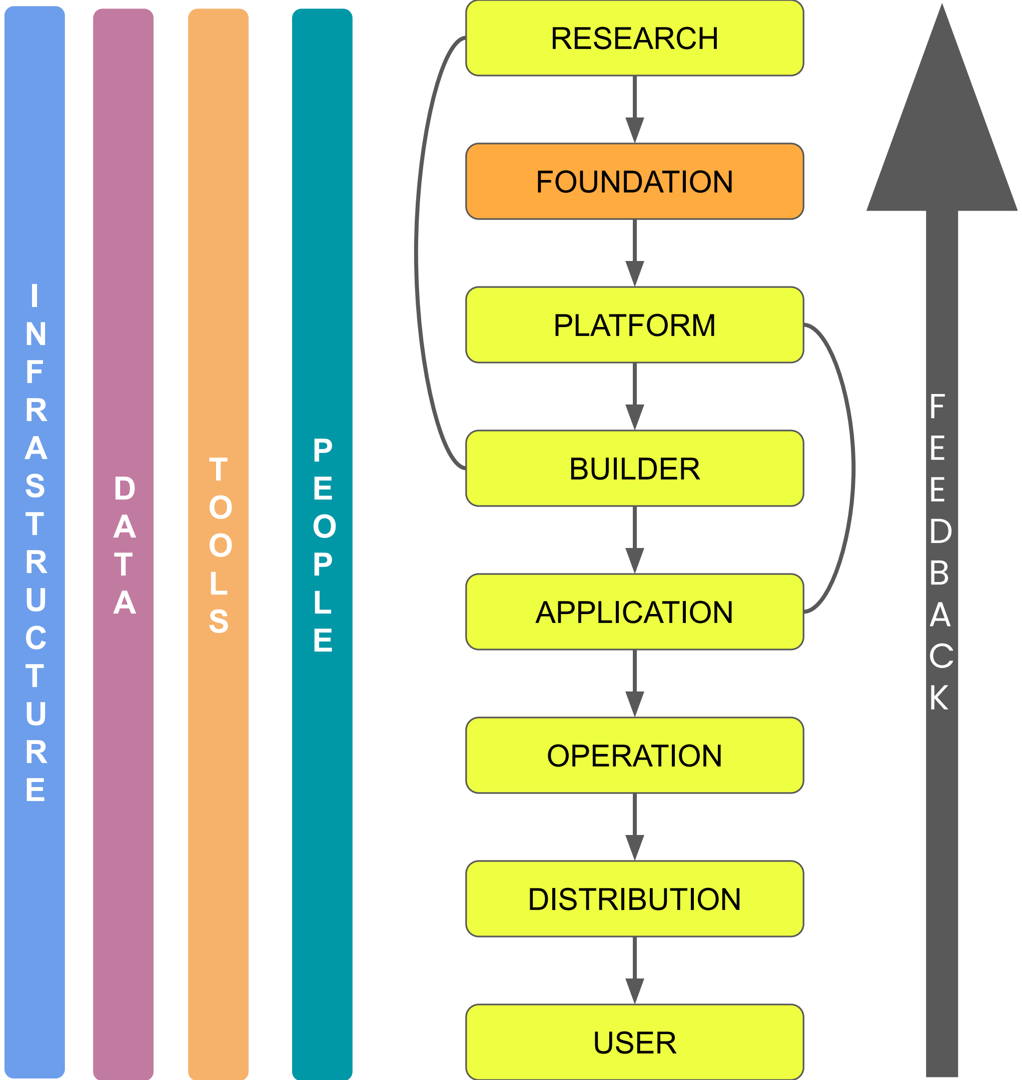
- **Web & Mobile Clients** – browsers, native apps, VS Code extensions, Slack bots
- **Prompt Optimization Platforms** – PromptPerfect, FlowGPT for refining prompts.



People

Research Layer

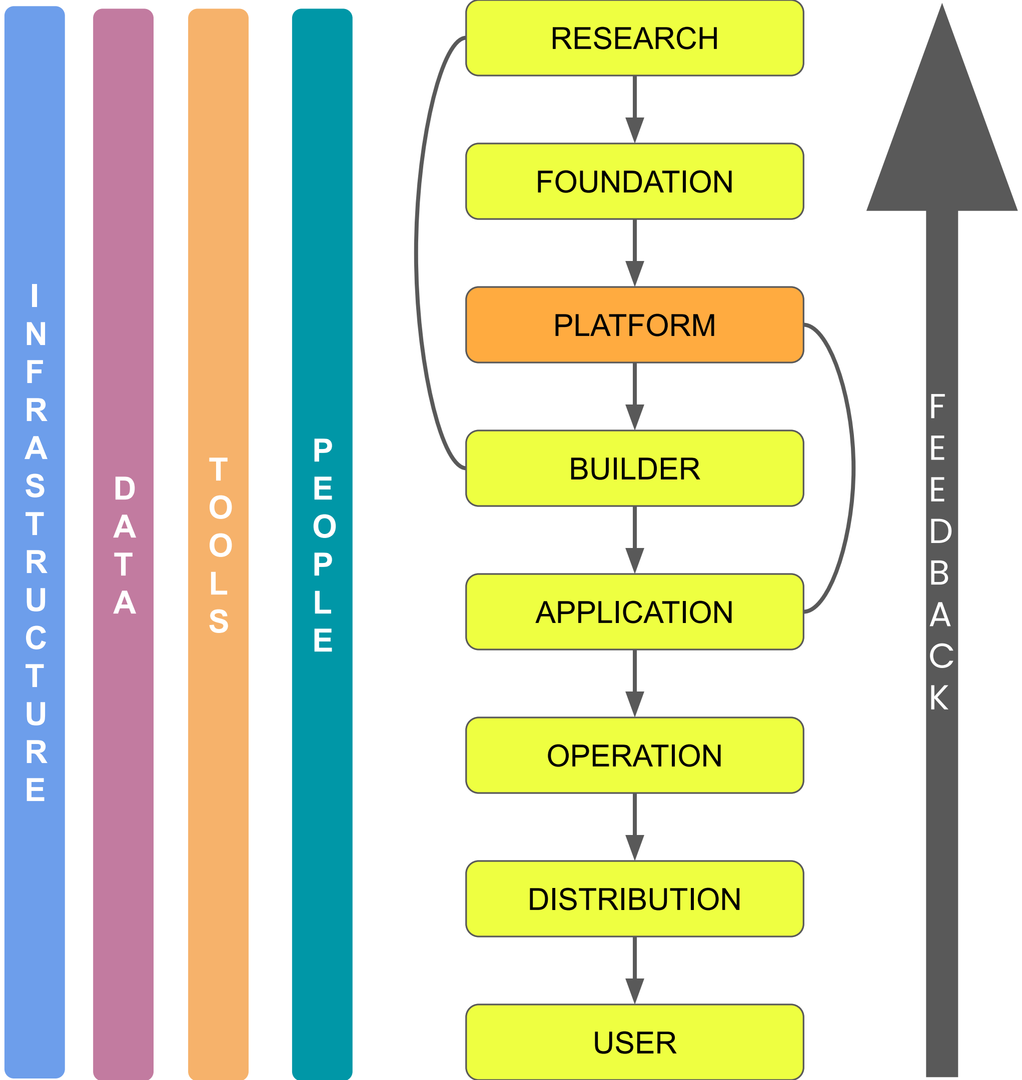
- **AI Researchers**– invent algorithms, architectures, and training methods.
- **Mathematicians / Statisticians** – formalize optimization, probability, and learning theories.
- **Research Engineers** – implement and test experimental ideas efficiently.
- **Graduate Students & Interns** – provide experimentation bandwidth and exploratory labor.



People

Foundation Layer

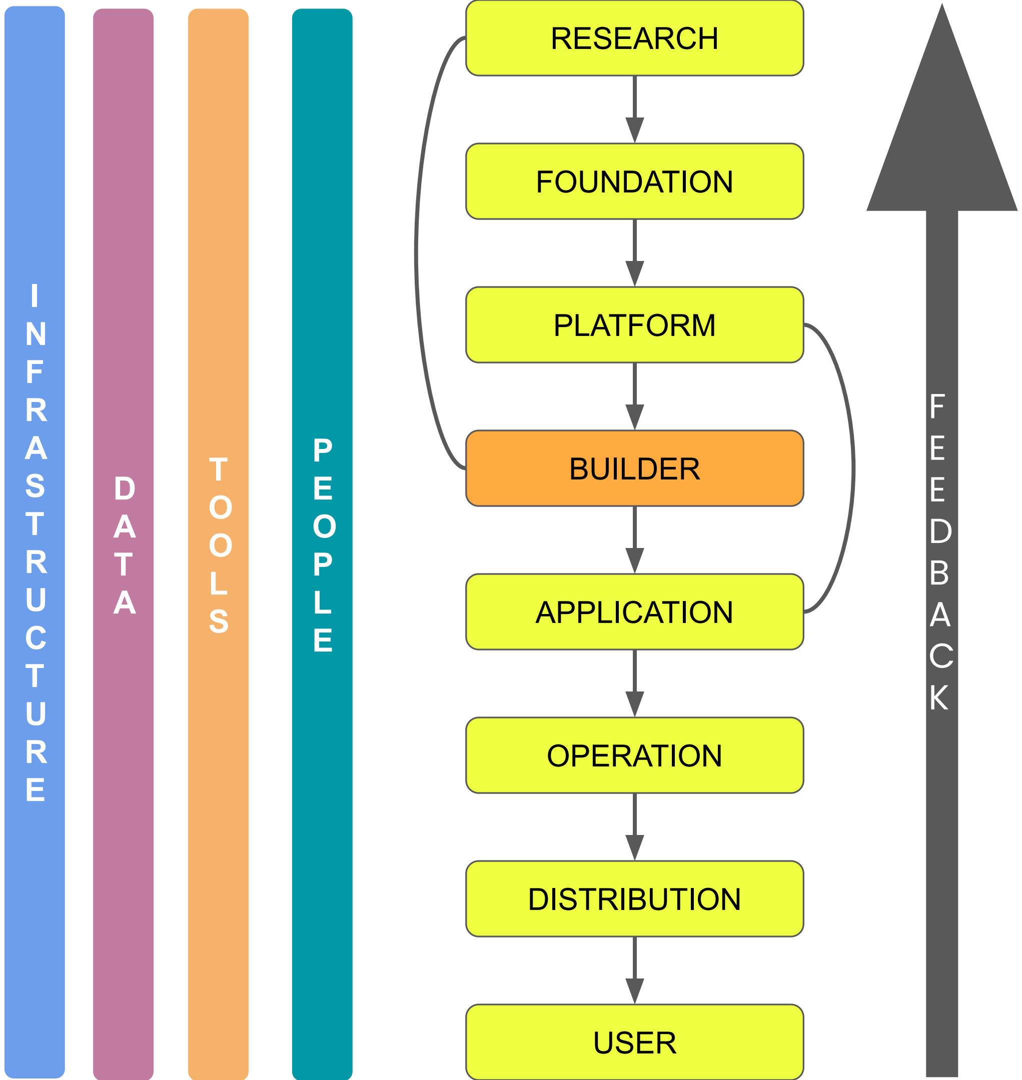
- **Applied Scientists** – translate research innovations into scalable architectures.
- **Data Engineers/Curators** – clean, deduplicate, and balance massive training datasets.
- **Distributed Systems Engineers** – optimize compute, storage, and networking for model training.
- **Infrastructure Architects** – design GPU/TPU clusters and storage for model training.



People

Platform Layer

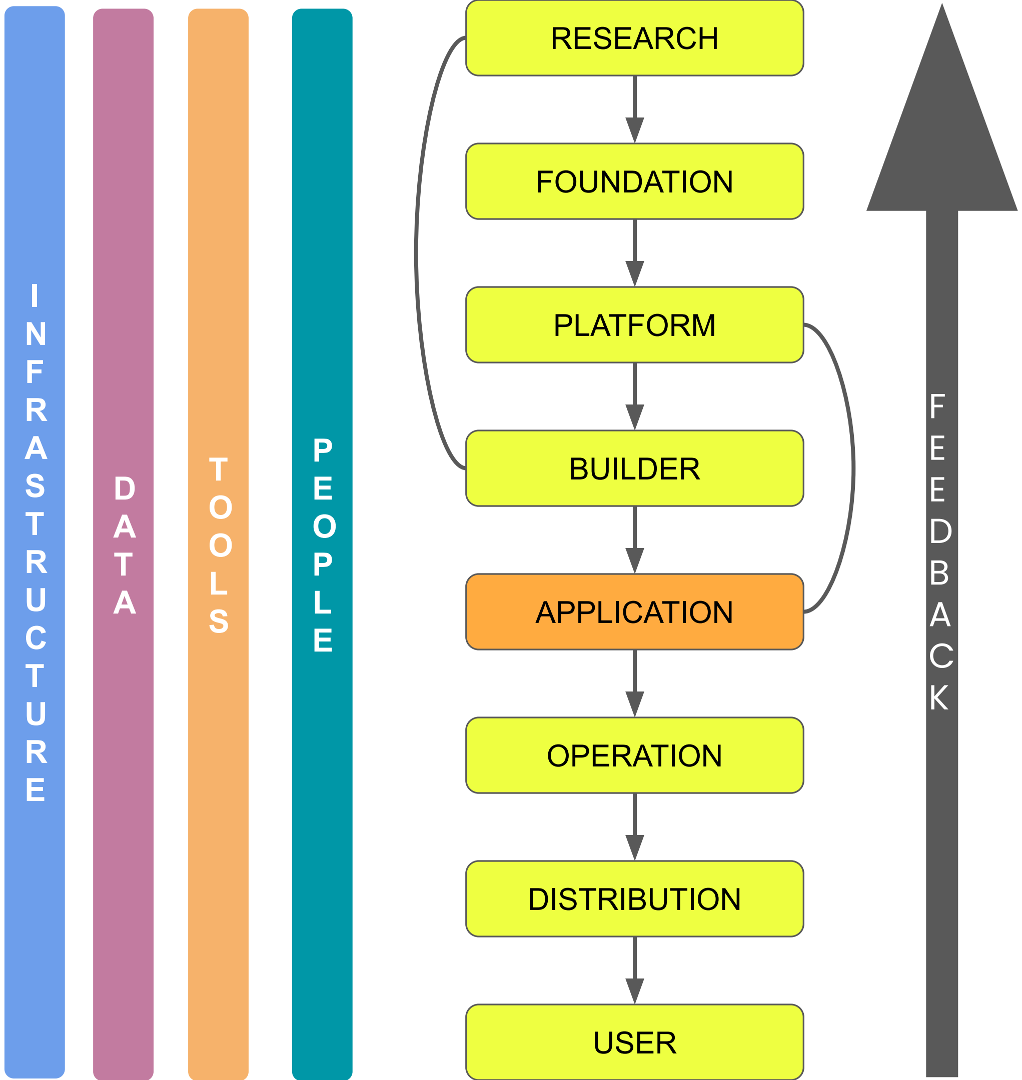
- **MLOps / DevOps Engineers** – handle model serving, containerization, and scaling.
- **Backend Developers** – build inference APIs and SDKs for model access.
- **Infrastructure Reliability Engineers** – ensure uptime, latency, and resilience.
- **Security Experts** – implement API safety, access control, and rate limiting.



People

Builder Layer

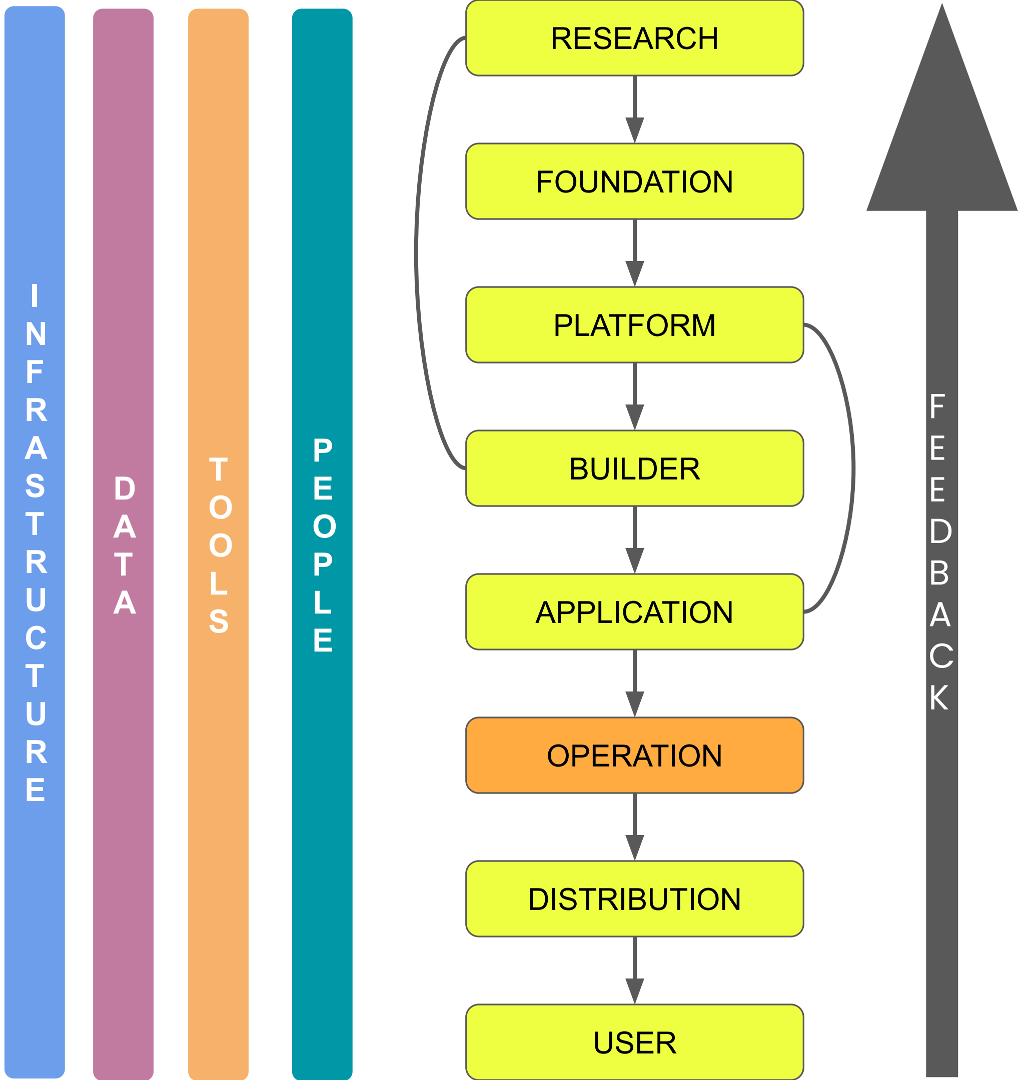
- **AI Engineers** – design reasoning chains, RAG systems, and agent loops.
- **Framework Developers** – build libraries like LangChain, LangGraph, or LlamaIndex.
- **Prompt Engineers** – craft and test prompt sequences for reliable reasoning.
- **Open-Source Contributors** – extend, document, and maintain builder-level frameworks.



People

Application Layer

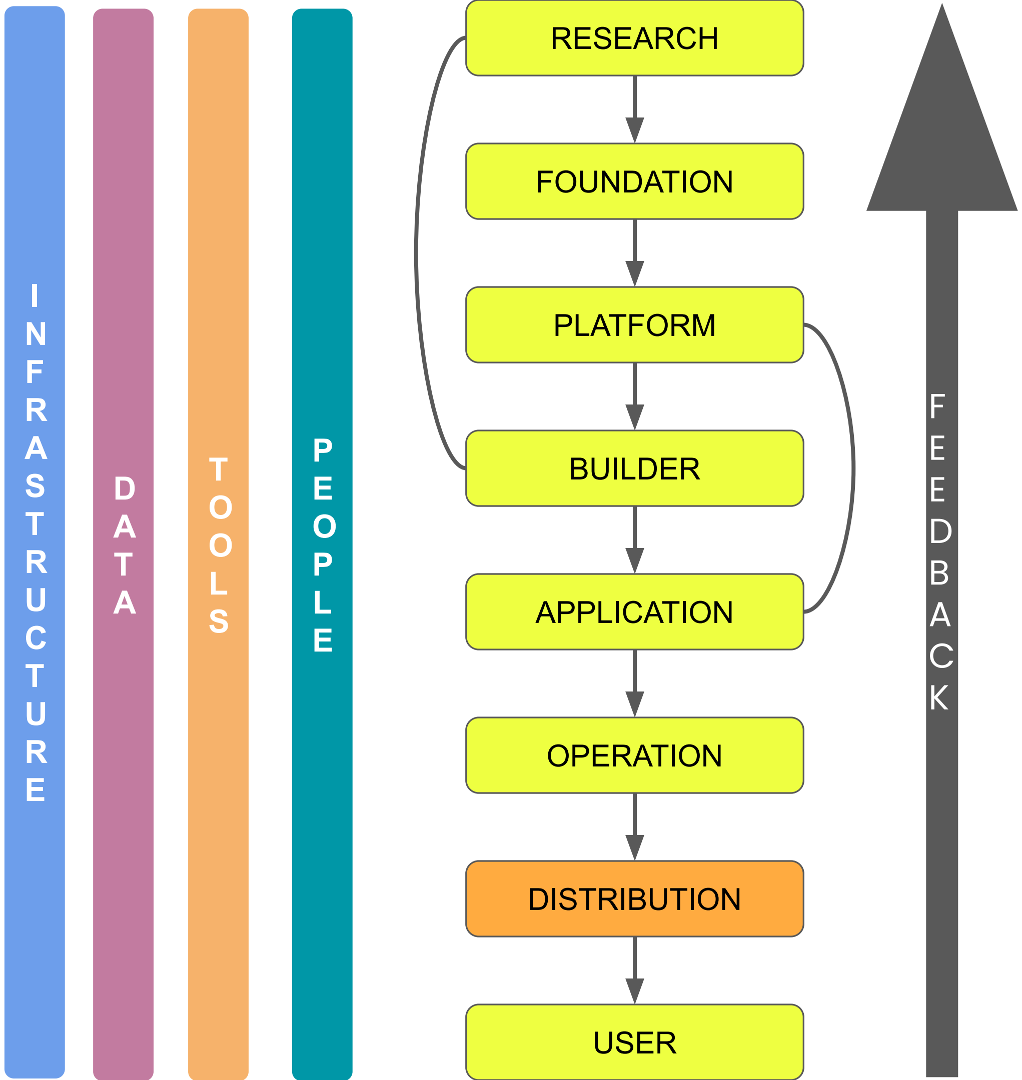
- **AI Engineers** – design reasoning chains, RAG systems, and agent loops.
- **Product Managers** – define use cases, target users, and success metrics.
- **Frontend & UX Designers** – design intuitive AI interfaces and visual feedback systems.
- **Full-Stack Developers** – integrate builder logic into functional products.



People

Operation Layer

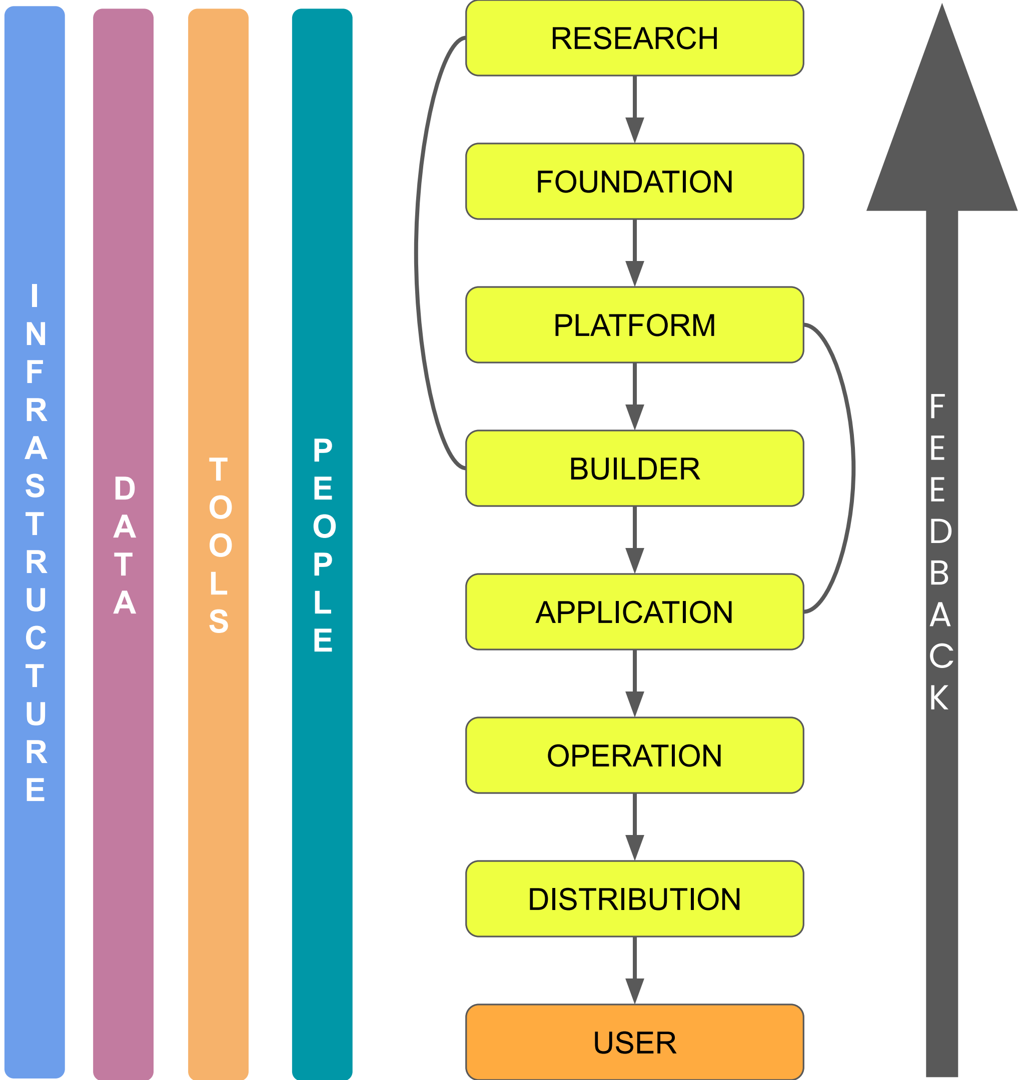
- **AIOps / LLMOps Engineers** – monitor model behavior and performance in production.
- **Observability Experts** – study logs and telemetry to detect drifts.
- **Trust & Safety Teams** – enforce output guardrails, content policies, and ethical standards.



People

Distribution Layer

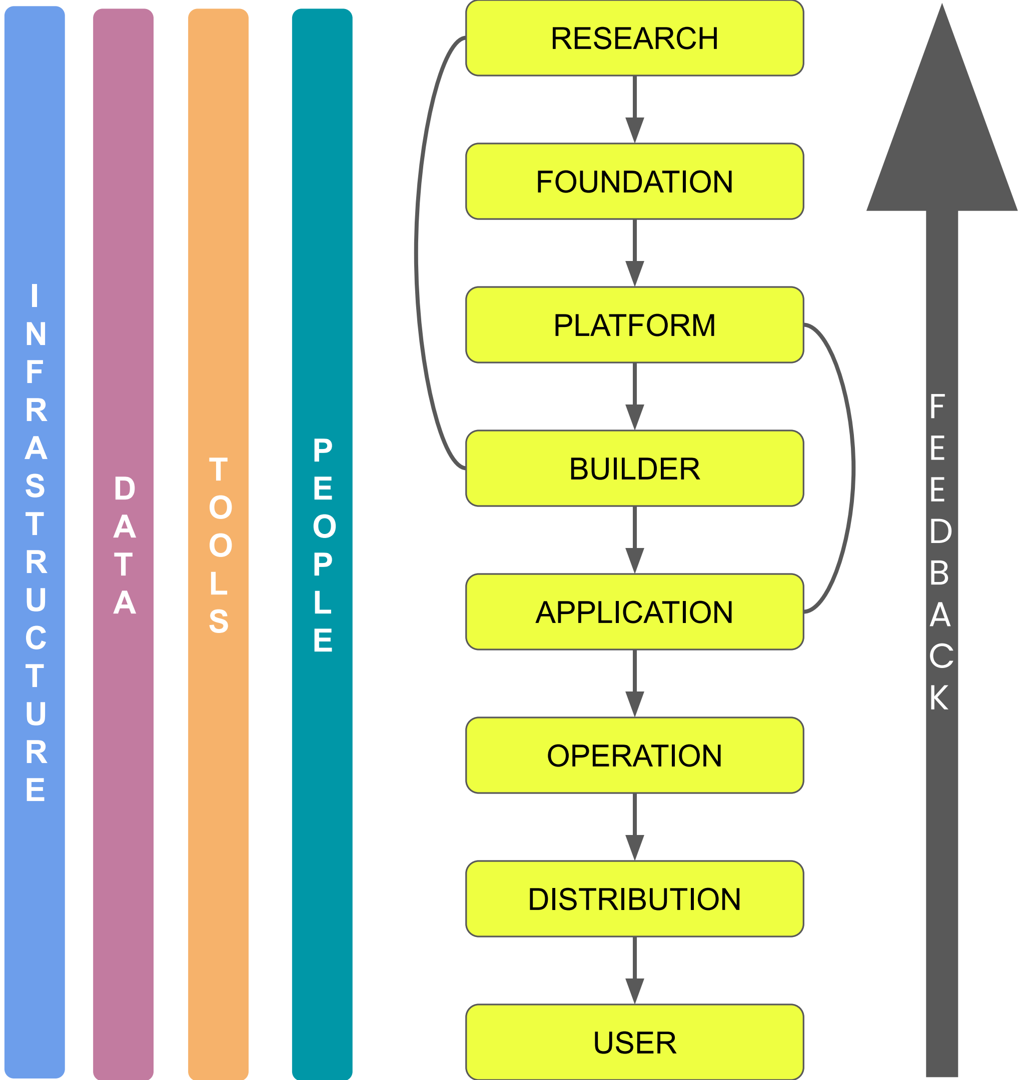
- **Growth & Product Marketers** – craft messaging, demos, and launch campaigns.
- **Developer Relations (DevRel) Teams** – build tutorials, docs, and SDKs for external developers.
- **Sales Engineers / Solution Architects** – customize deployments for enterprise clients.
- **Localization Specialists** – adapt AI products for different languages and regions.



People

User Layer

- **End Users / Consumers** – interact with AI systems through prompts, queries, and feedback.
- **Enterprise Users / Teams** – integrate AI into workflows and business processes.
- **Prompt Creators / Community Builders** – share templates, custom GPTs, and extensions.
- **Educators** – teach others how to use AI effectively and responsibly.



What have we achieved?

- **Full picture of GenAI**
- **Fit any term**
- **Feel Literate**
- **Fit yourself - create your own roadmap**

Jaate Jaate!

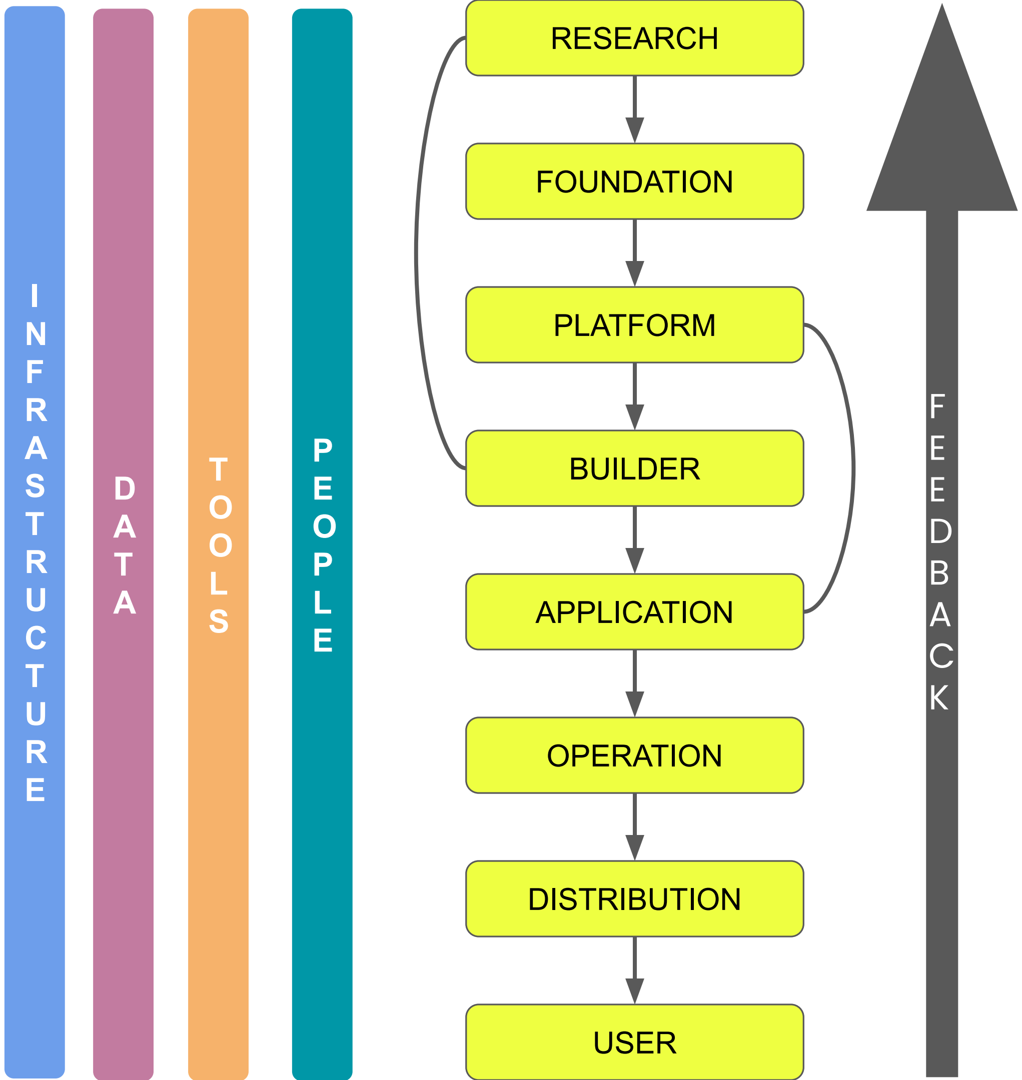
How to test this map

Working on a detailed resource

How am I feeling?

Map of GenAI

QUIZ



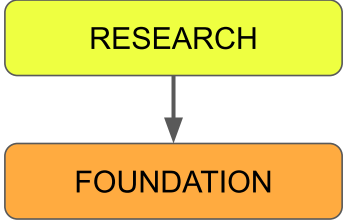
Agenda of the Live Session

- **Recap**
- **Test this Map**
- **Learn about new GenAI terms**
- **Interaction**

RESEARCH

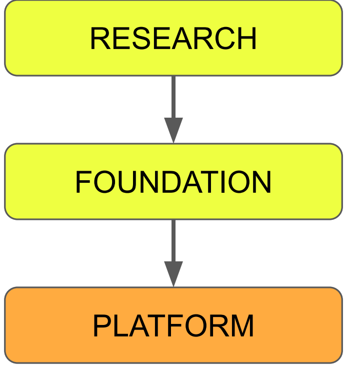
Research Layer

The birthplace of core AI innovations



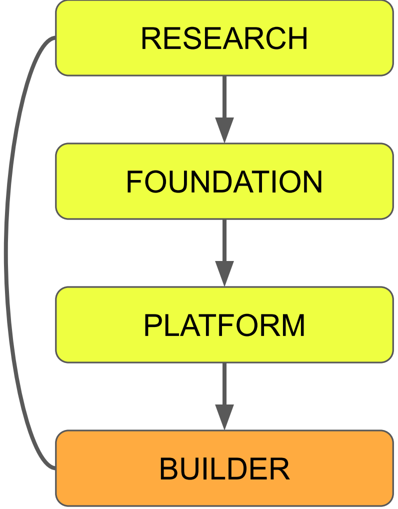
Foundation Layer

This is where research ideas are converted to large scale AI models



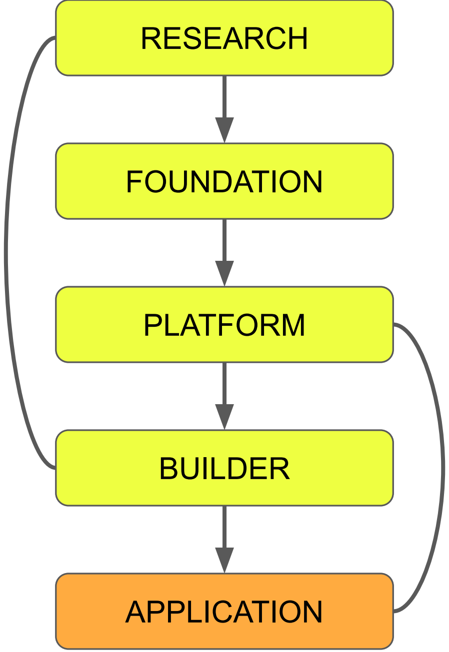
Platform Layer

This layer brings foundation models online



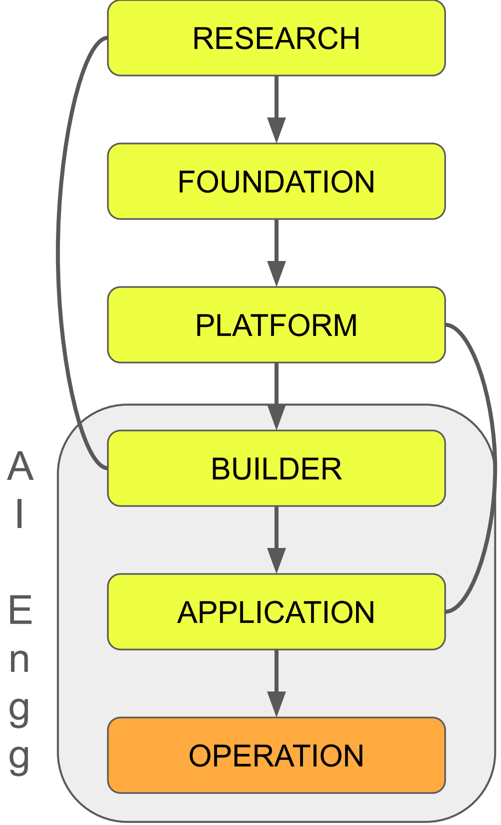
[Builder Layer]

This is where intelligence takes shape into usable systems.



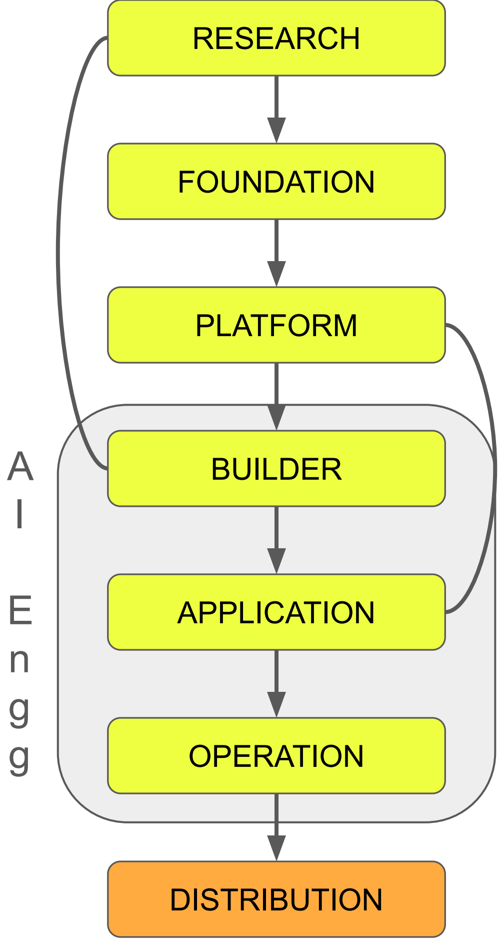
Application Layer

This is where AI capabilities are packaged into end user products



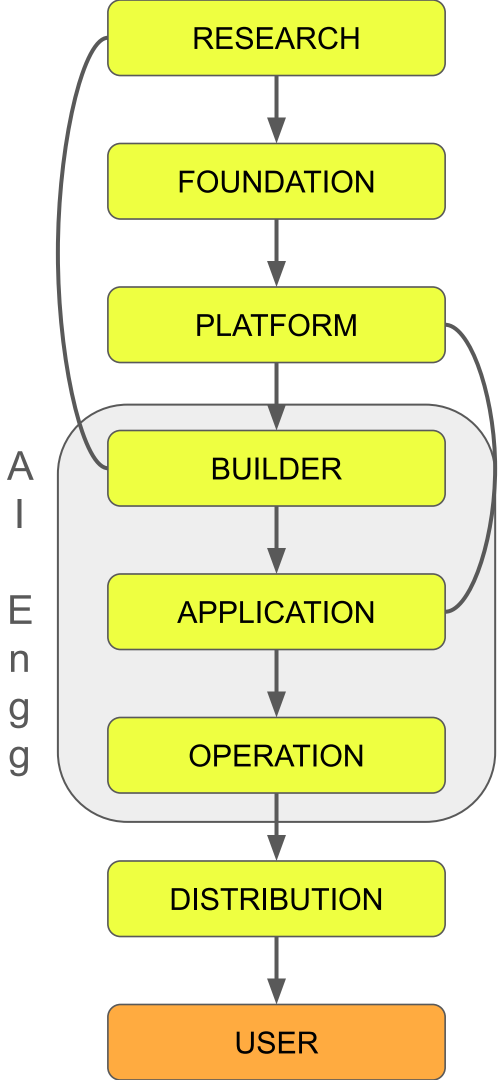
Operation Layer

This is where AI systems are deployed, monitored, and managed in production.



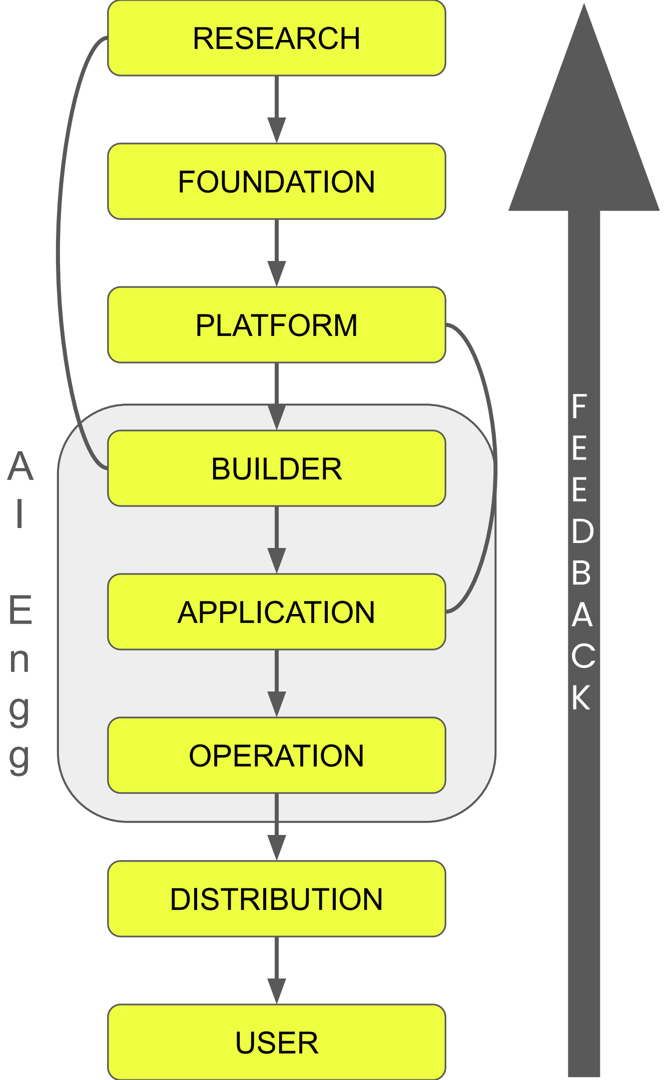
Distribution Layer

This is where AI products are reaches users through channels, platforms, and marketplaces.



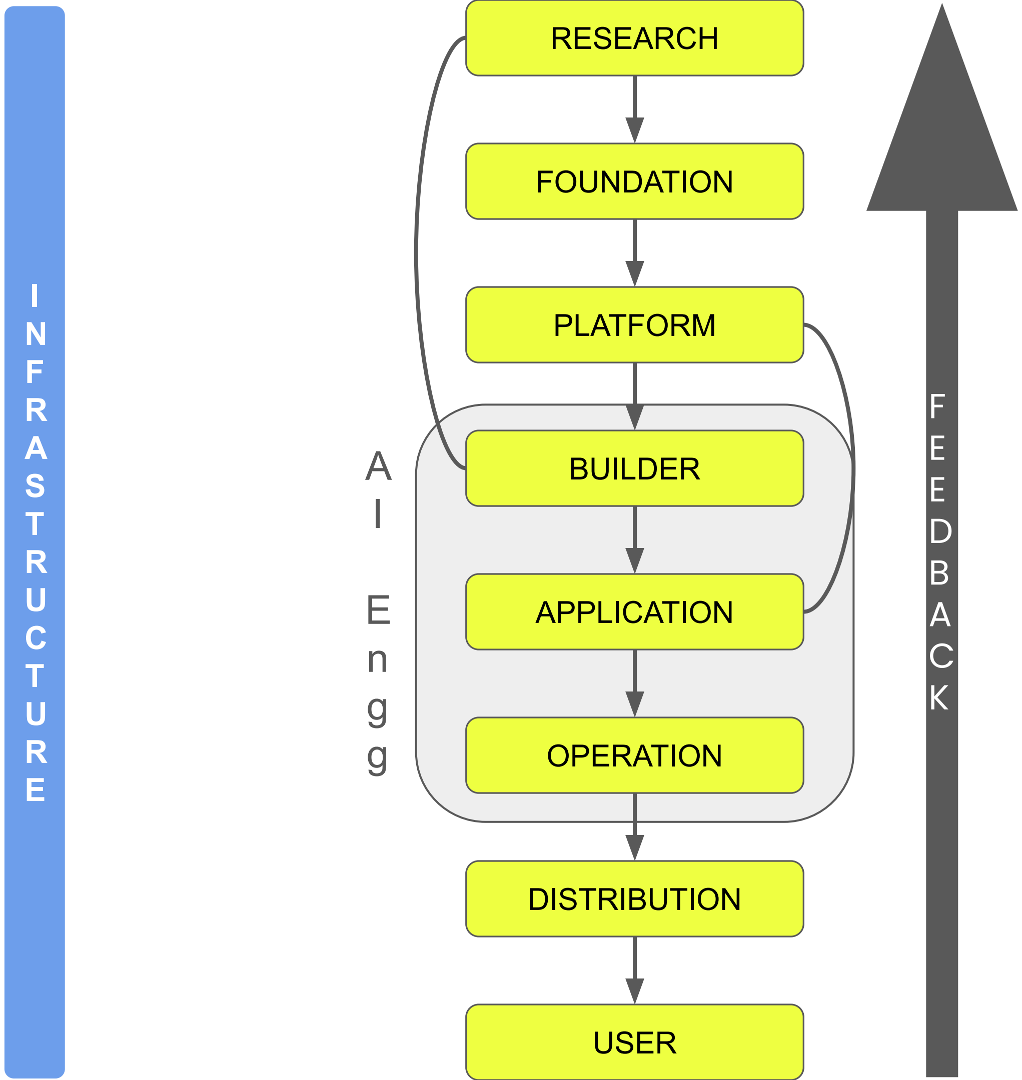
User Layer

The user layer is where users interact with AI



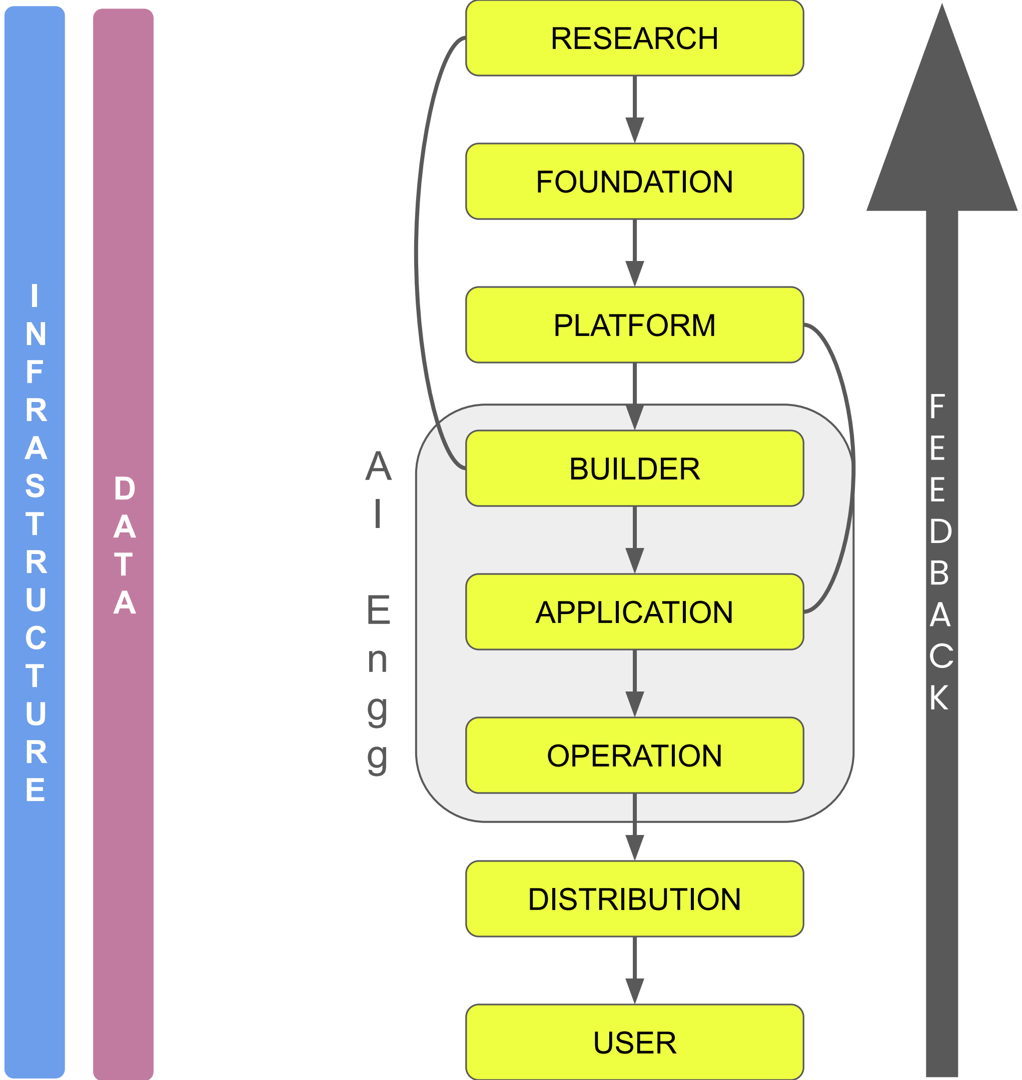
Feedback

The flow is bidirectional with a strong feedback loop going from user to research layer



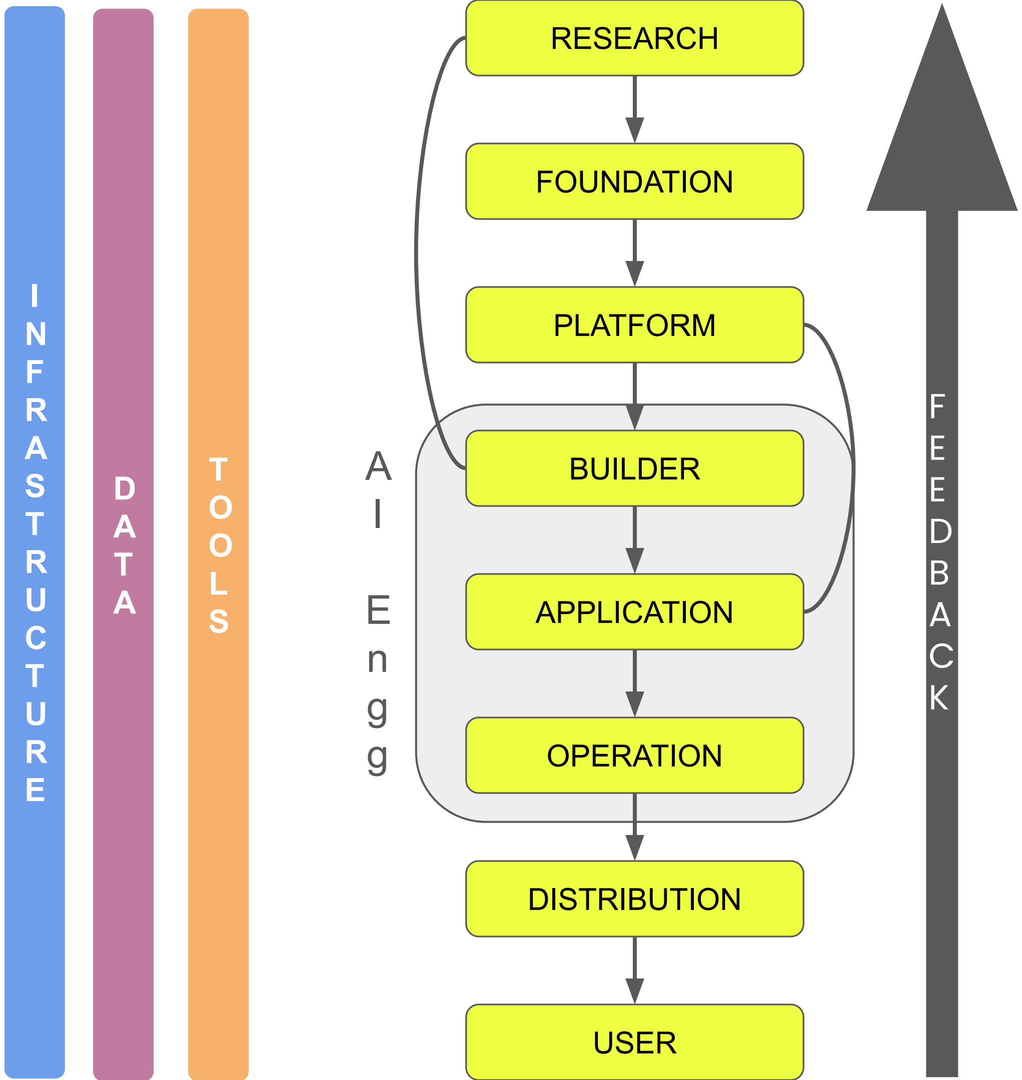
Infrastructure

Compute, networking, storage, and hardware required to train and serve models.



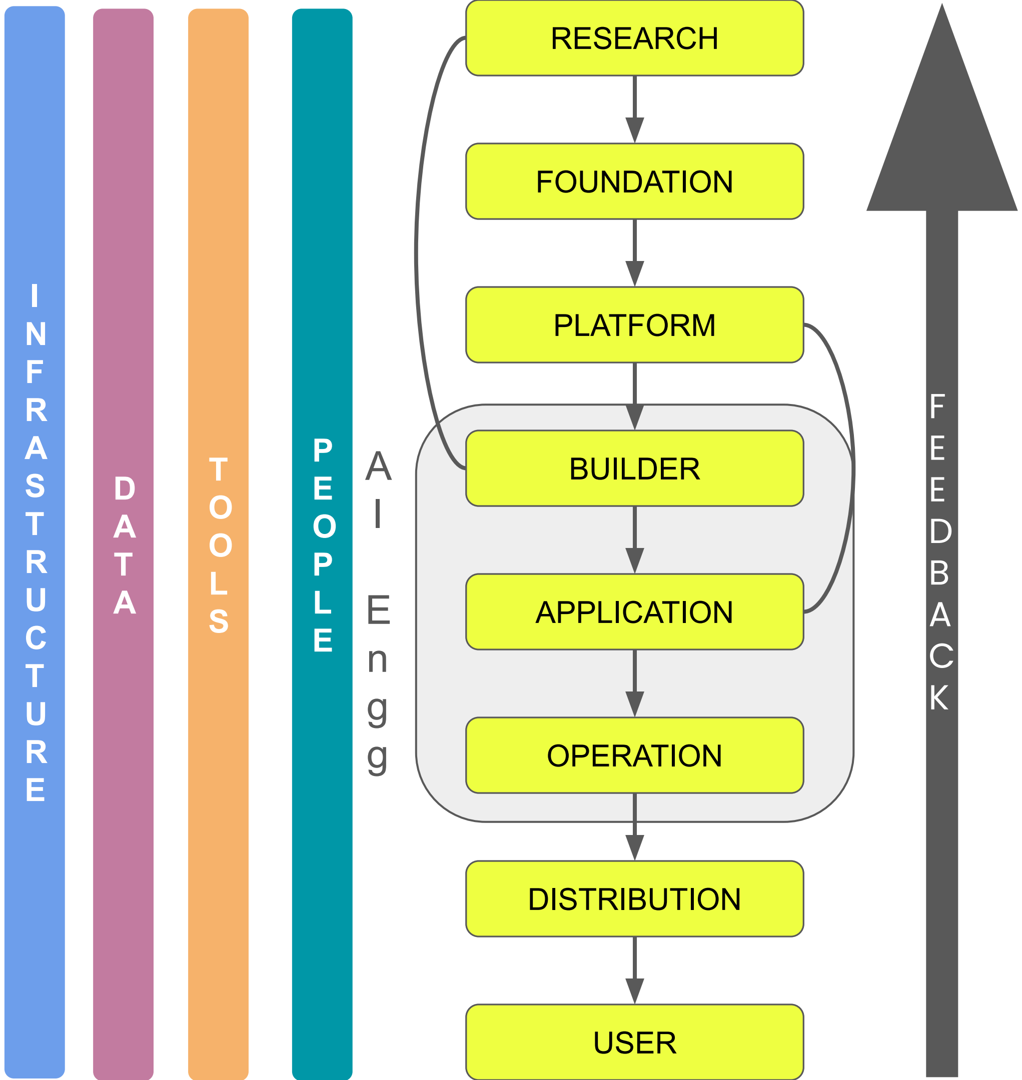
Data

The raw material for training, fine-tuning, evaluation, and operations.



Tools

Frameworks, libraries, runtimes, SDKs, and systems used to build, fine-tune, deploy, or orchestrate AI.



People

Researchers, engineers, builders, operators, and users who shape the ecosystem.